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(54) **SIC PLANAR MOSFETS WITH IMPROVED PERFORMANCE STRUCTURES**

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H01L 29/16	(2006.01)
H01L 29/423	(2006.01)

(52) **U.S. Cl.**

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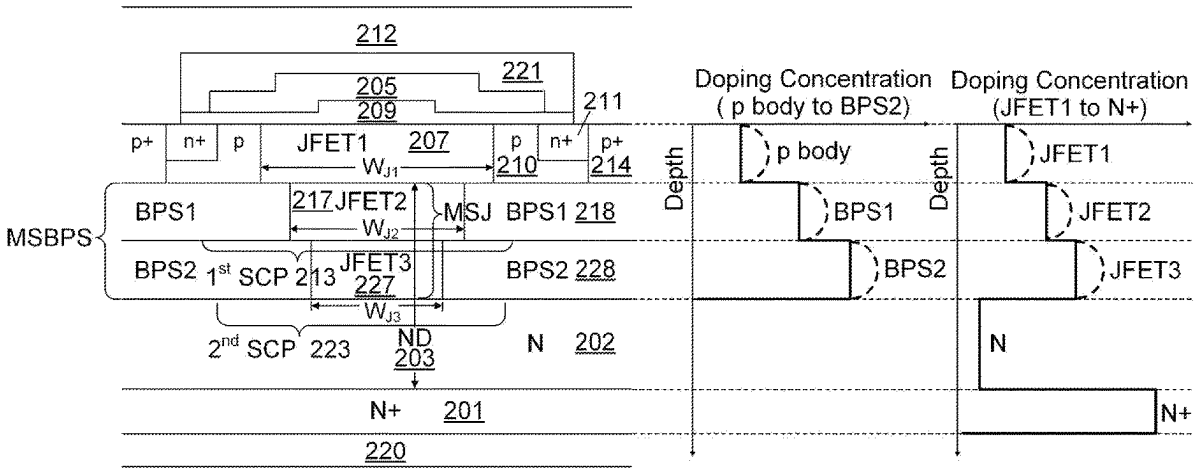
(57) **ABSTRACT**

An improved silicon carbide (SiC) planar MOSFET having at least one buried P-shield (BPS) region including one center portion and two side portions for the gate oxide electric field and saturation current reductions is disclosed. Two saturation current pitching (SCP) structures formed in two Junction Field Effect Transistor (JFET) regions sandwiched between the side portions and the center portion of the BPS region limit a saturation current of the device in a forward conduction stage for the short-circuit capability improvement. Moreover, a Junction barrier Schottky diode (JBSD) is integrated with the SiC MOSFET in a location between the two adjacent split gate electrodes for the reverse conduction switching loss reduction.



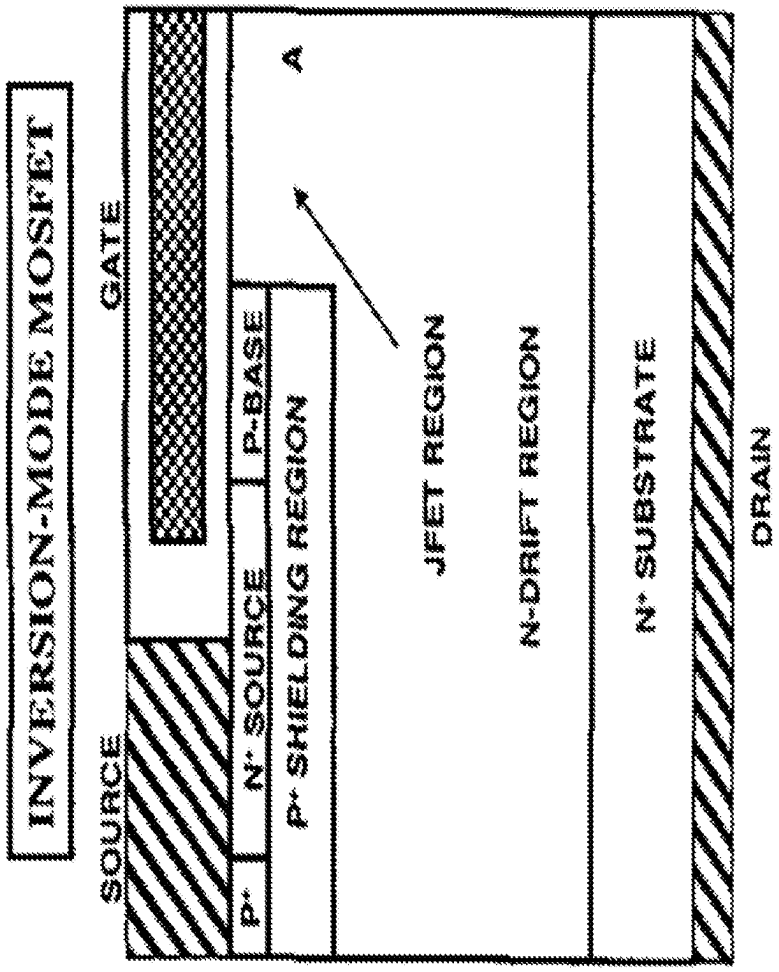


Fig. 1 (Prior Art)

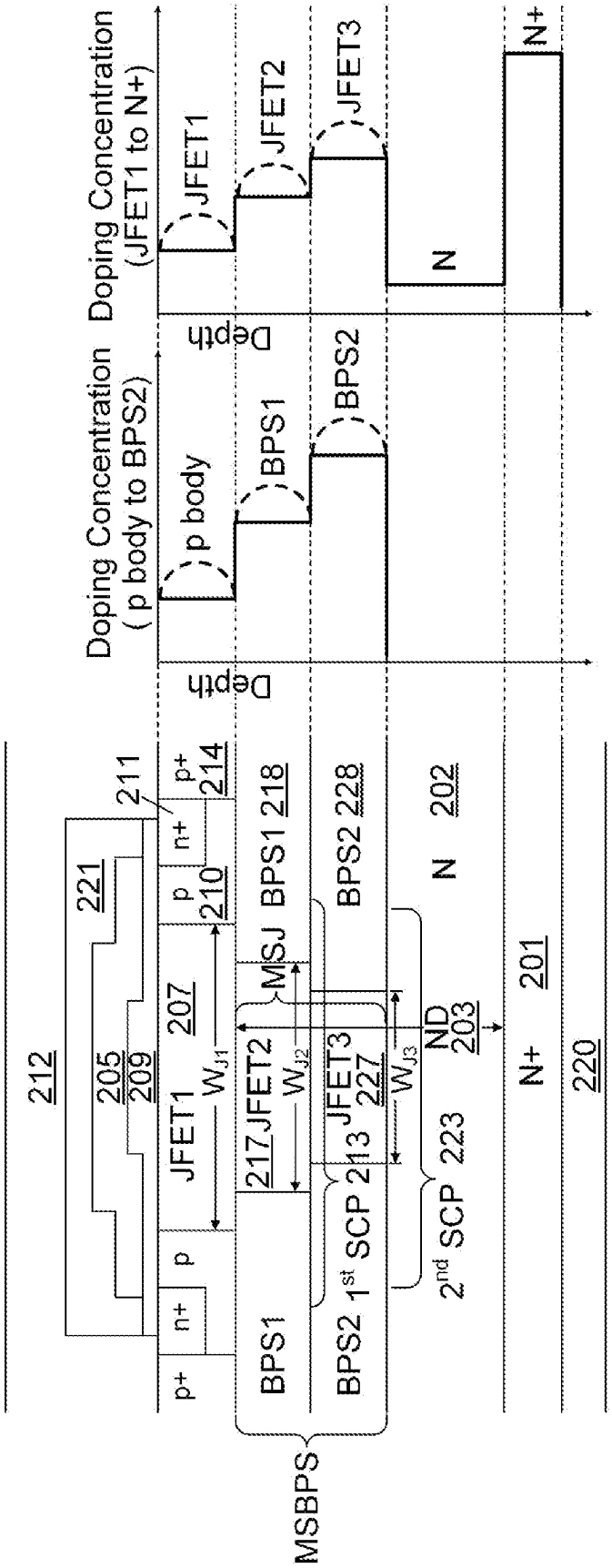

Doping Concentration
(p body to BPS2)
Doping Concentration
(JFET1 to N+)
Depth
p body
BPS1
BPS2
JFET1
JFET2
JFET3
N
N+
212
221
205
209
211
p+
n+
p
JFET1
207
p
210
n+
p+
214
217
JFET2
MSJ
BPS1 218
BPS2 228
N 202
ND
203
JFET3
227
213
BPS1
BPS2
1st SCP
2nd SCP 223
N+ 201
220
MSBPS

Fig. 2A

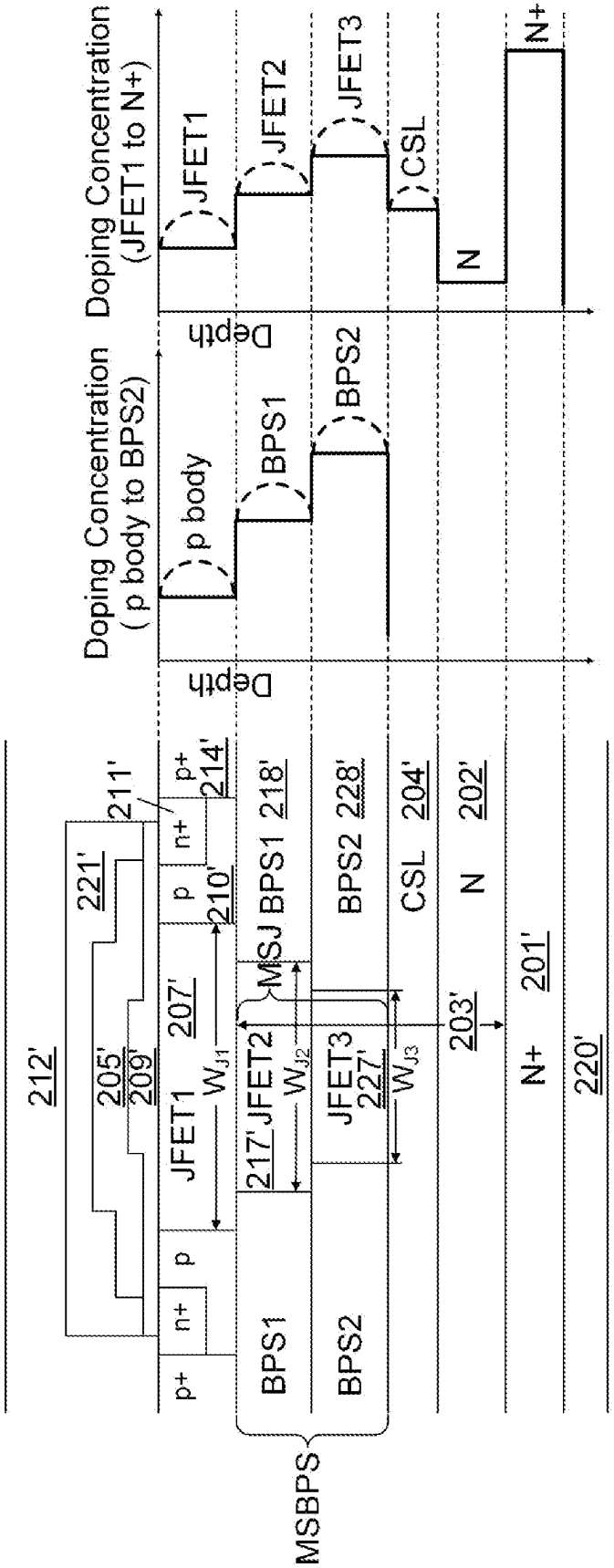

Doping Concentration
(JFET1 to N+)
Doping Concentration
(p body to BPS2)
Depth
Depth
JFET1
JFET2
JFET3
CSL
N
N+
p body
BPS1
BPS2
212'
221'
205'
209'
211'
214'
210'
207'
217'
218'
228'
227'
204'
202'
203'
201'
220'
p+
n+
p
JFET1
JFET2
JFET3
MSJ
BPS1
BPS2
CSL
N
N+
MSBPS

Fig. 2B

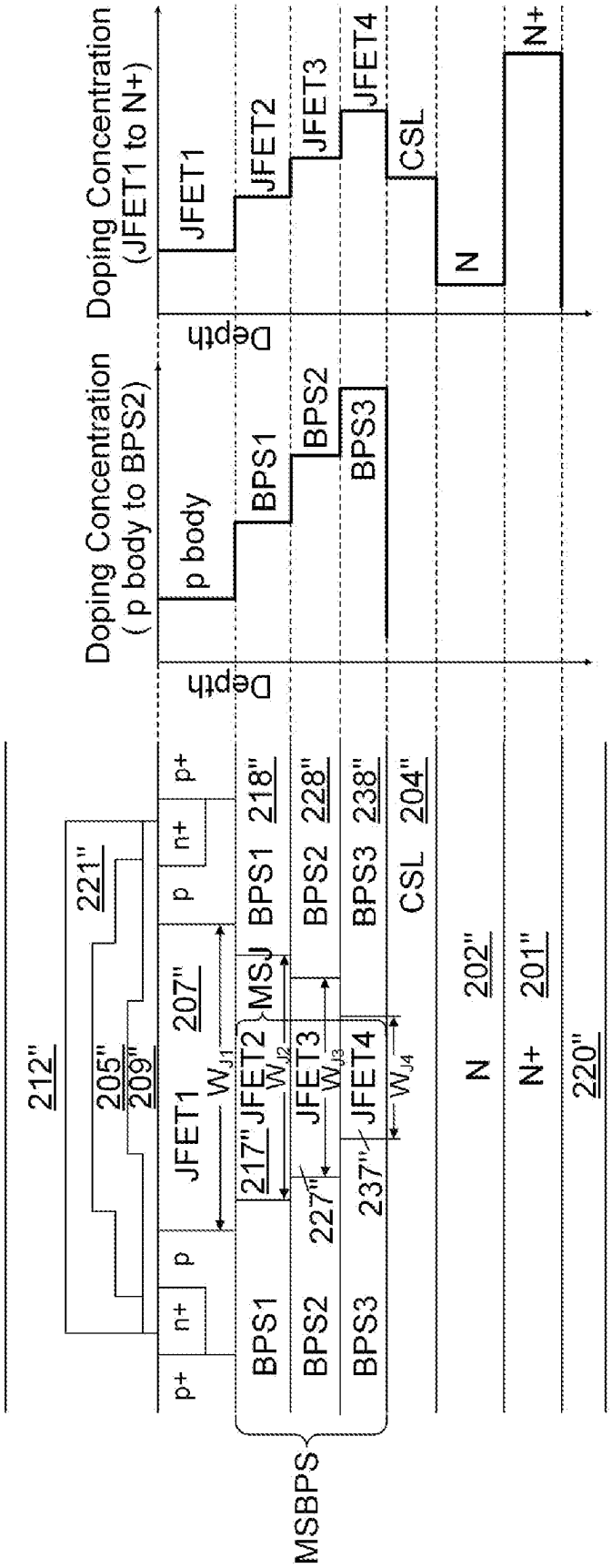

212"
221"
205"
209"
207"
p+
n+
p
JFET1
W_{J1}
217"
JFET2
W_{J2}
MSJ
JFET3
W_{J3}
227"
237"
JFET4
W_{J4}
BPS1
BPS2
BPS3
218"
228"
238"
CSL
204"
N
202"
N+
201"
220"
MSBPS
Depth
Doping Concentration
(p body to BPS2)
p body
BPS1
BPS2
BPS3
Depth
Doping Concentration
(JFET1 to N+)
JFET1
JFET2
JFET3
JFET4
CSL
N
N+

Fig. 2C

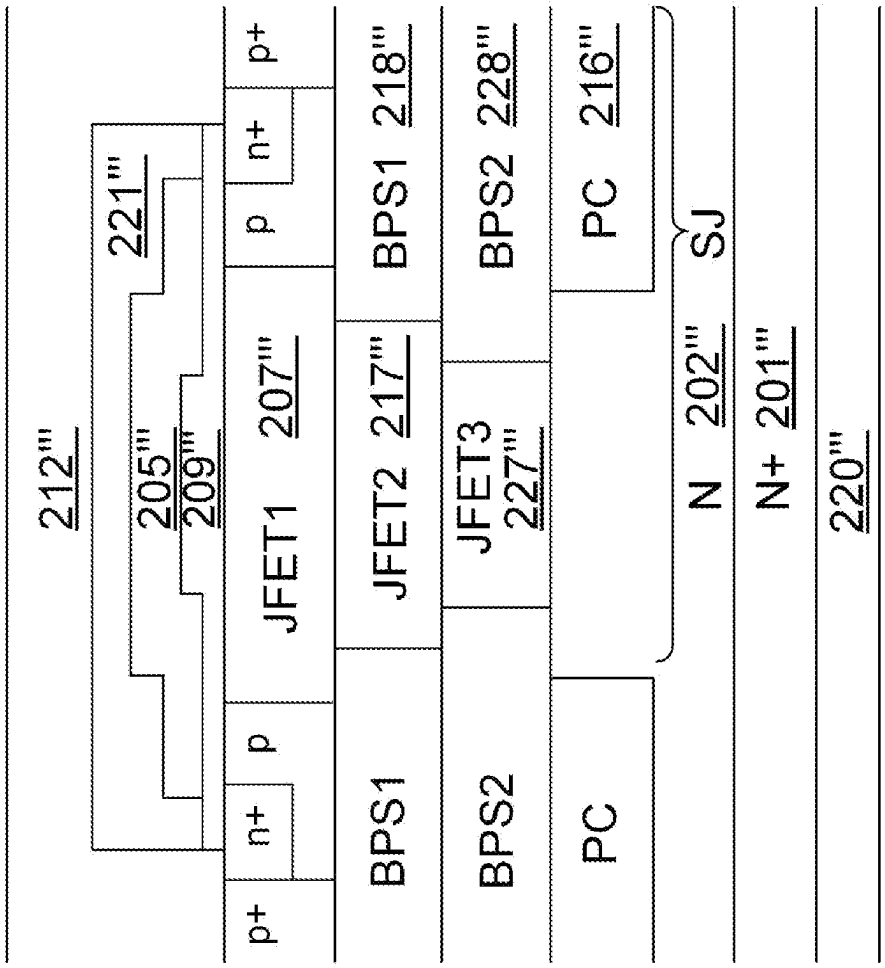

212'''
221'''
205'''
209'''
p+
n+
p
JFET1 207'''
p
n+
p+
BPS1
JFET2 217'''
BPS1 218'''
BPS2
JFET3 227'''
BPS2 228'''
PC
PC 216'''
N 202'''
SJ
N+ 201'''
220'''

Fig. 2D

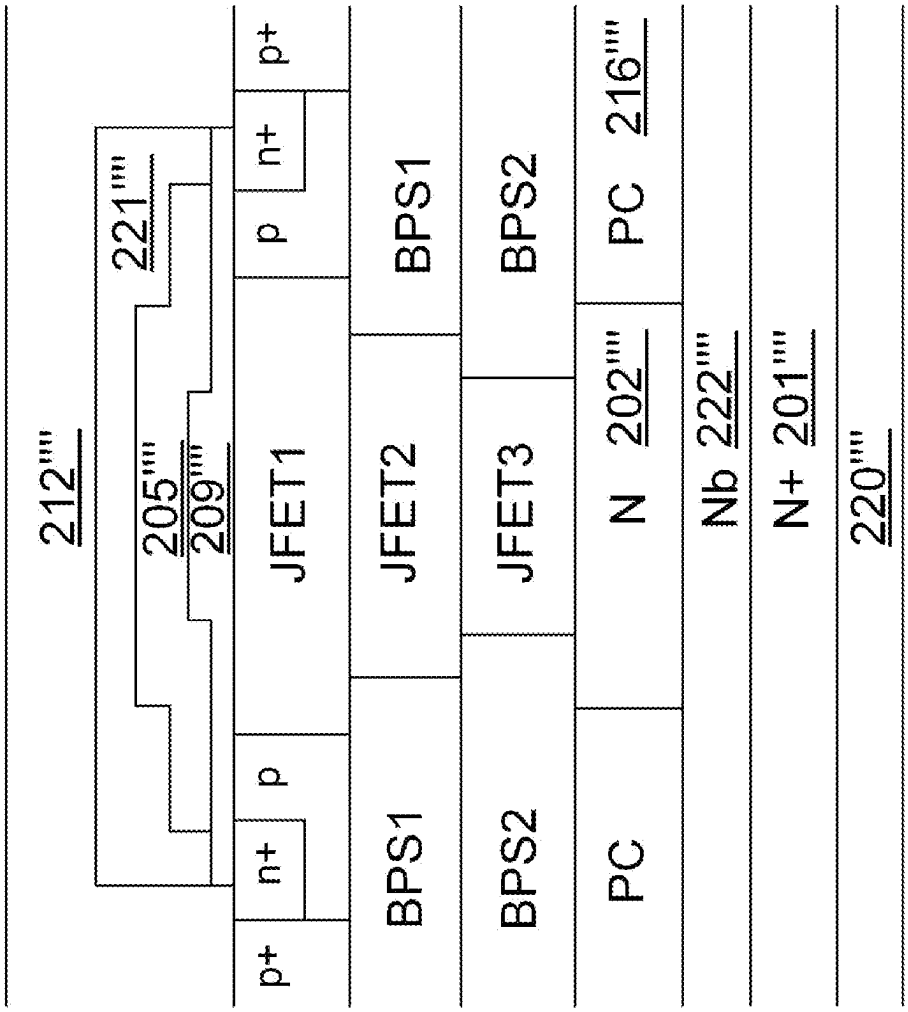

212''''
221''''
205''''
209''''
p+
n+
p
JFET1
p
n+
p+
BPS1
JFET2
BPS1
BPS2
JFET3
BPS2
PC
N 202''''
PC 216''''
Nb 222''''
N+ 201''''
220''''

Fig. 2E

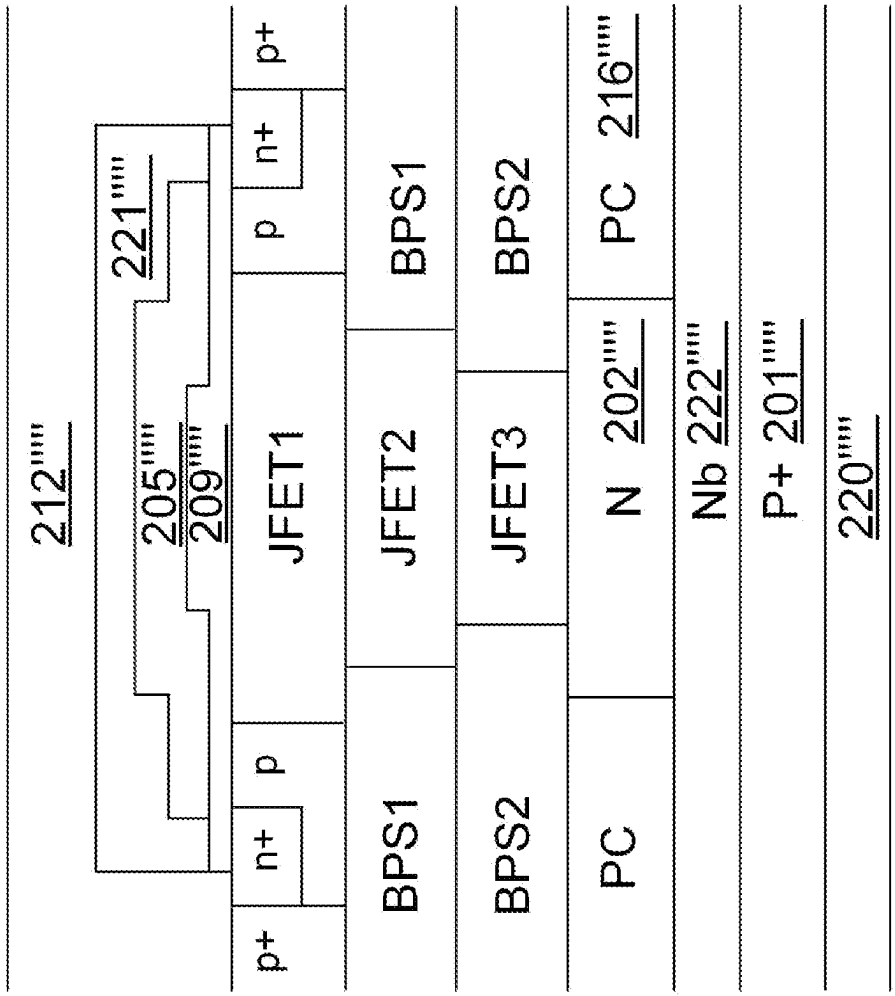

212''''
205''''
209''''
221''''
p+
n+
p
JFET1
p
n+
p+
BPS1
JFET2
BPS1
BPS2
JFET3
BPS2
PC
N 202''''
PC 216''''
Nb 222''''
P+ 201''''
220''''

Fig. 2F

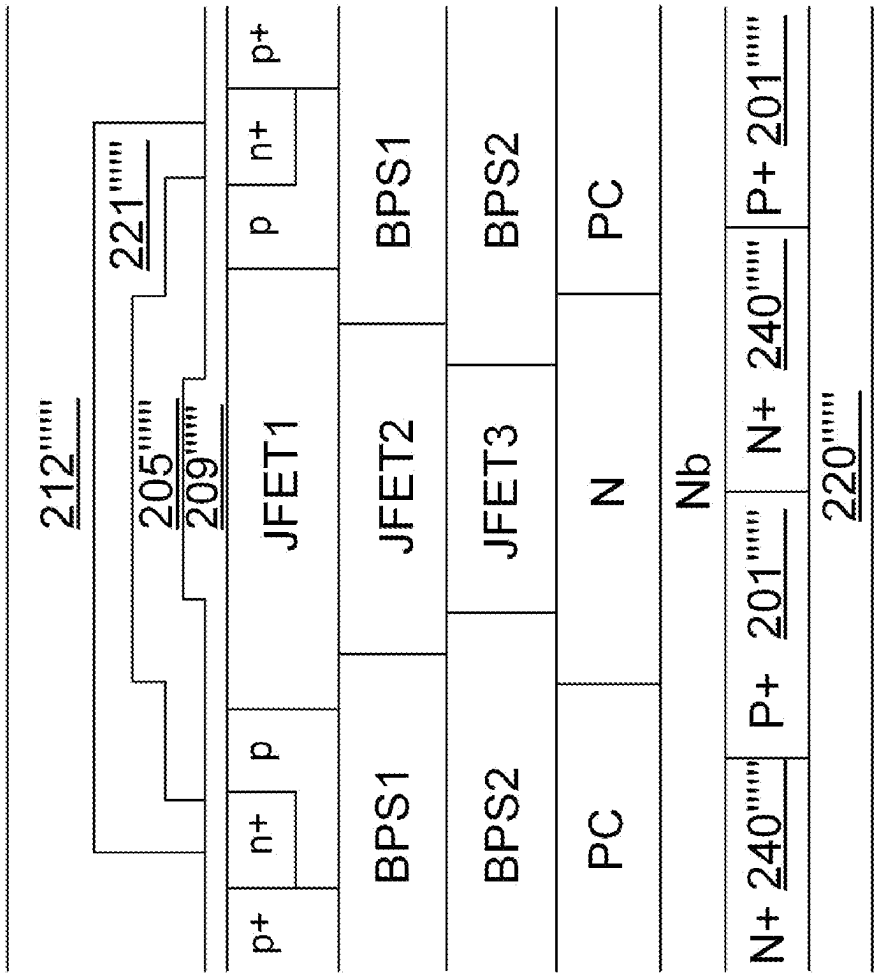

212''''''
221''''''
205''''''
209''''''
p+
n+
p
JFET1
p
n+
p+
BPS1
JFET2
BPS2
JFET3
BPS2
PC
N
PC
Nb
N+ 240''''''
P+ 201''''''
N+ 240''''''
P+ 201''''''
220''''''

Fig. 2G

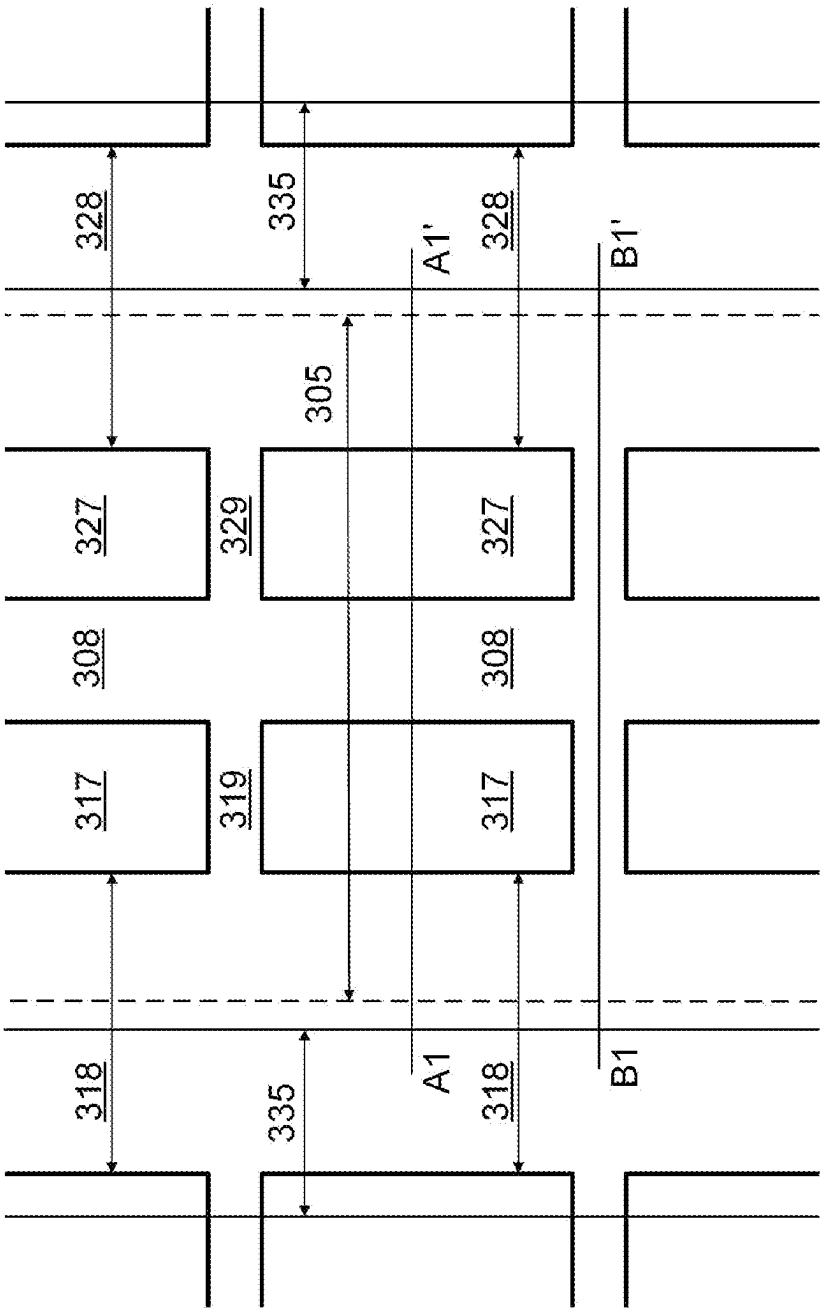

328
335
A1'
328
B1'
305
327
329
327
308
308
317
319
317
318
335
A1
318
B1

Fig. 3A

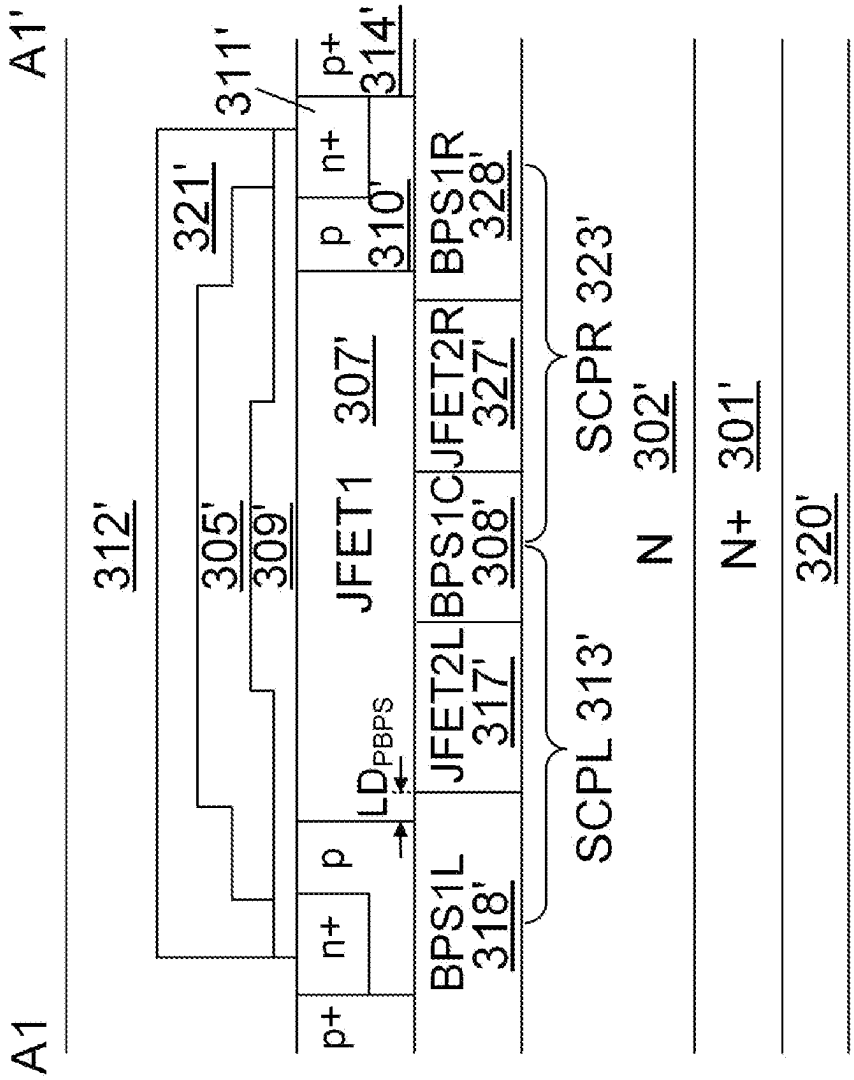

A1
A1'
312'
321'
305'
309'
311'
p+
n+
p
LD$_{PBPS}$
JFET1
307'
p
n+
p+
310'
314'
BPS1L
318'
JFET2L
317'
BPS1C
308'
JFET2R
327'
BPS1R
328'
SCPL 313'
SCPR 323'
N
302'
N+
301'
320'

Fig. 3B

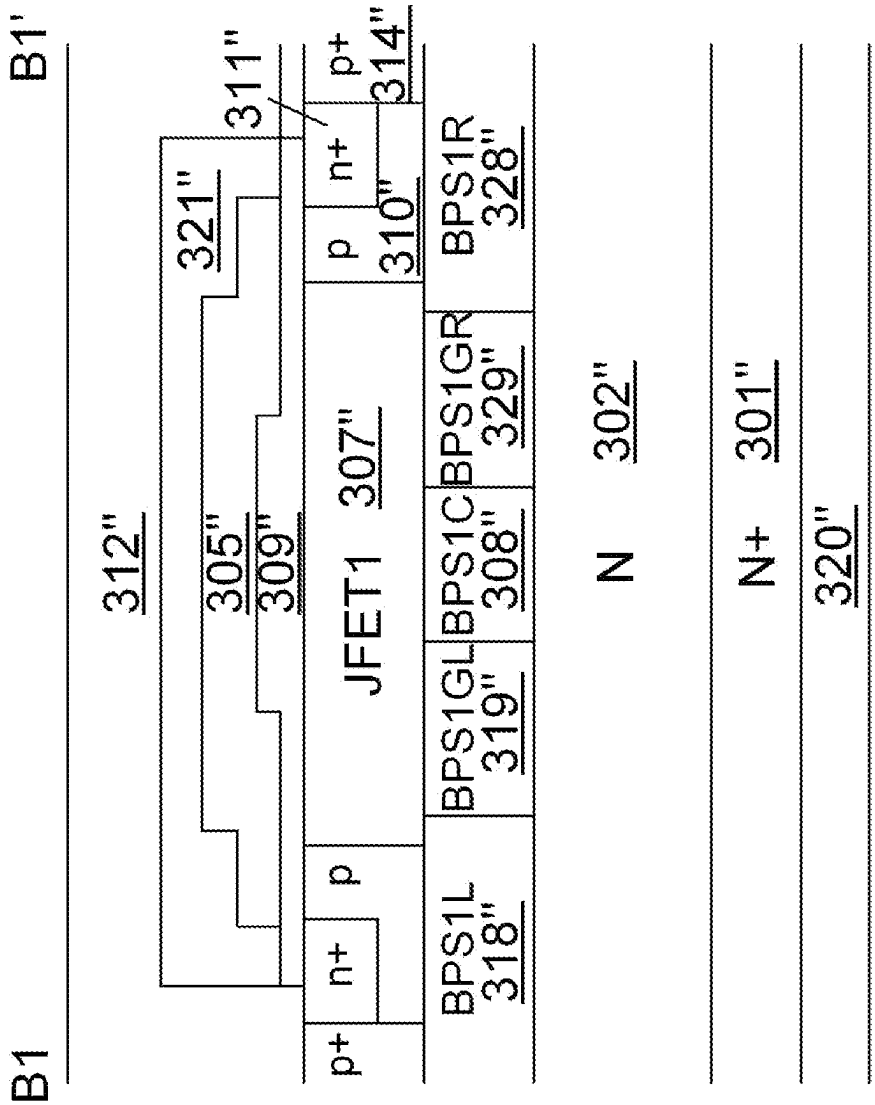

B1
B1'
312"
305"
309"
321"
311"
p+
n+
p
JFET1 307"
p
n+
p+
314"
310"
BPS1L 318"
BPS1GL 319"
BPS1C 308"
BPS1GR 329"
BPS1R 328"
N 302"
N+ 301"
320"

Fig. 3C

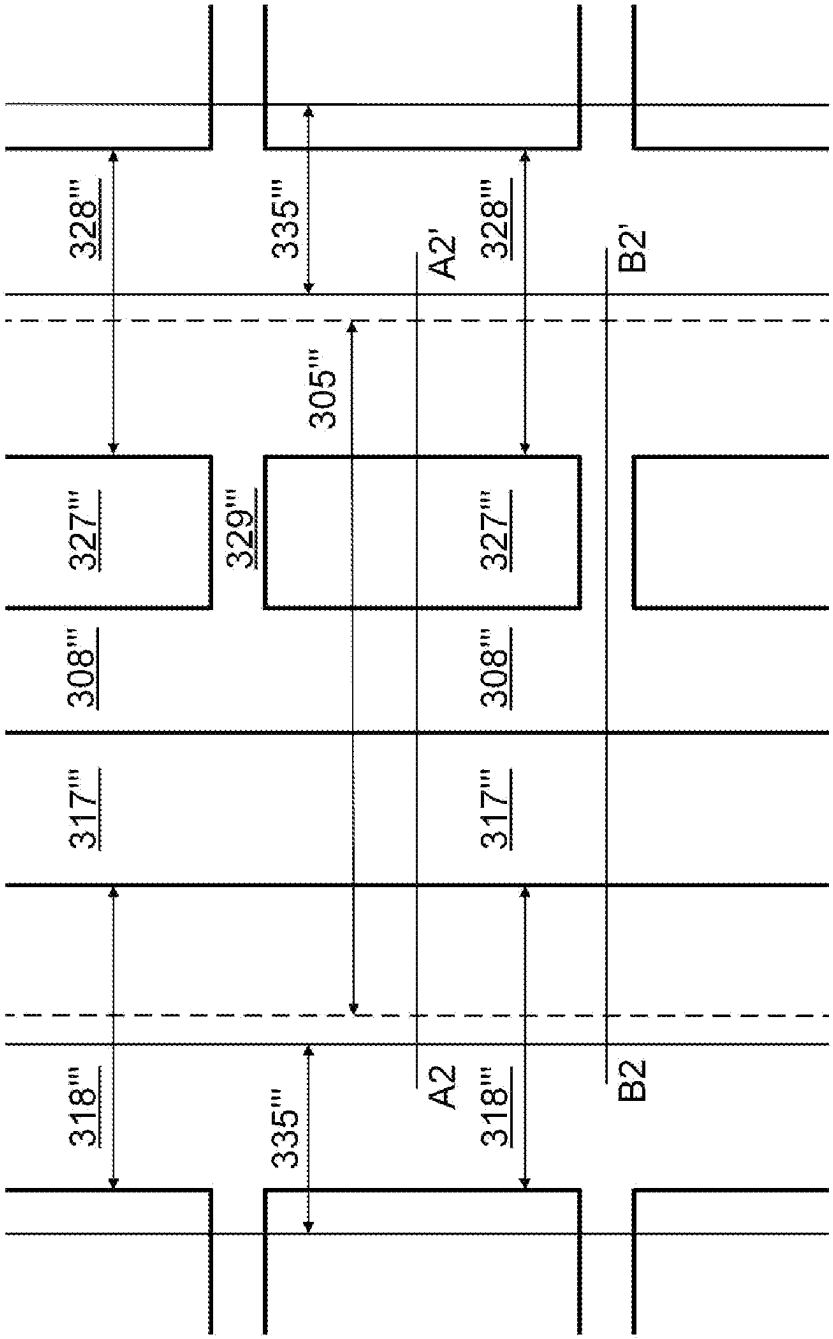

328'''
335'''
A2'
328'''
B2'
305'''
327'''
329'''
327'''
308'''
308'''
317'''
317'''
318'''
335'''
A2
318'''
B2

Fig. 3D

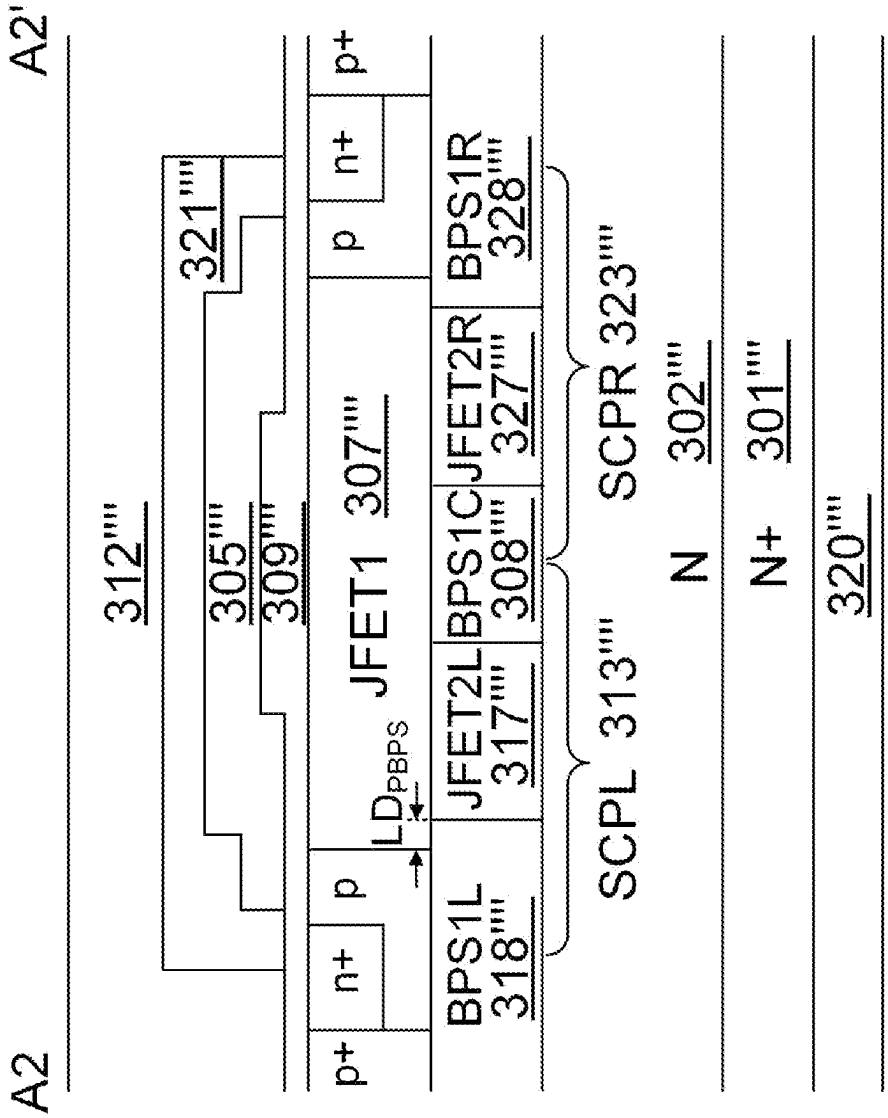

A2
A2'
312''''
321''''
305''''
309''''
p+
n+
p
LDPBPS
JFET1 307''''
p
n+
p+
BPS1L 318''''
JFET2L 317''''
BPS1C 308''''
JFET2R 327''''
BPS1R 328''''
SCPL 313''''
SCPR 323''''
N 302''''
N+ 301''''
320''''

Fig. 3E

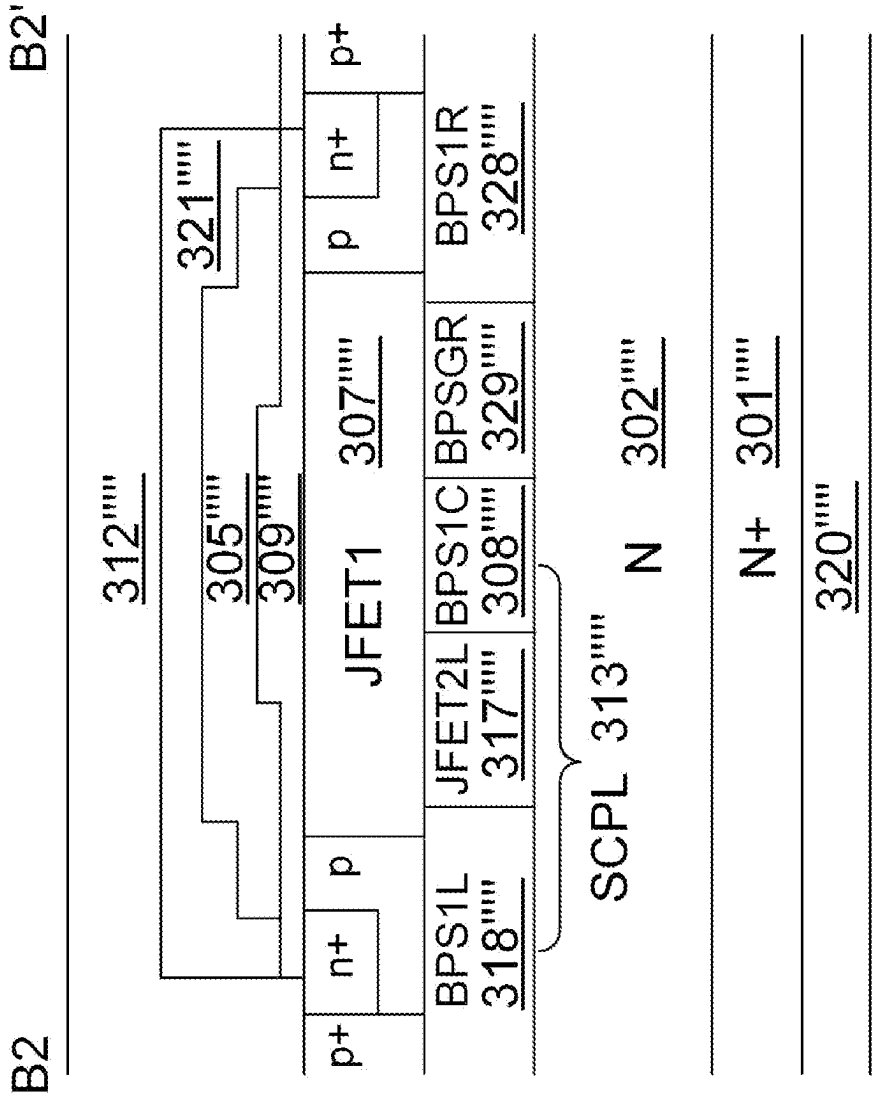

B2
B2'
312''''
321''''
305''''
309''''
p+
n+
p
JFET1
307''''
p
n+
p+
BPS1L
318''''
JFET2L
317''''
BPS1C
308''''
BPSGR
329''''
BPS1R
328''''
SCPL 313''''
N 302''''
N+ 301''''
320''''

Fig. 3F

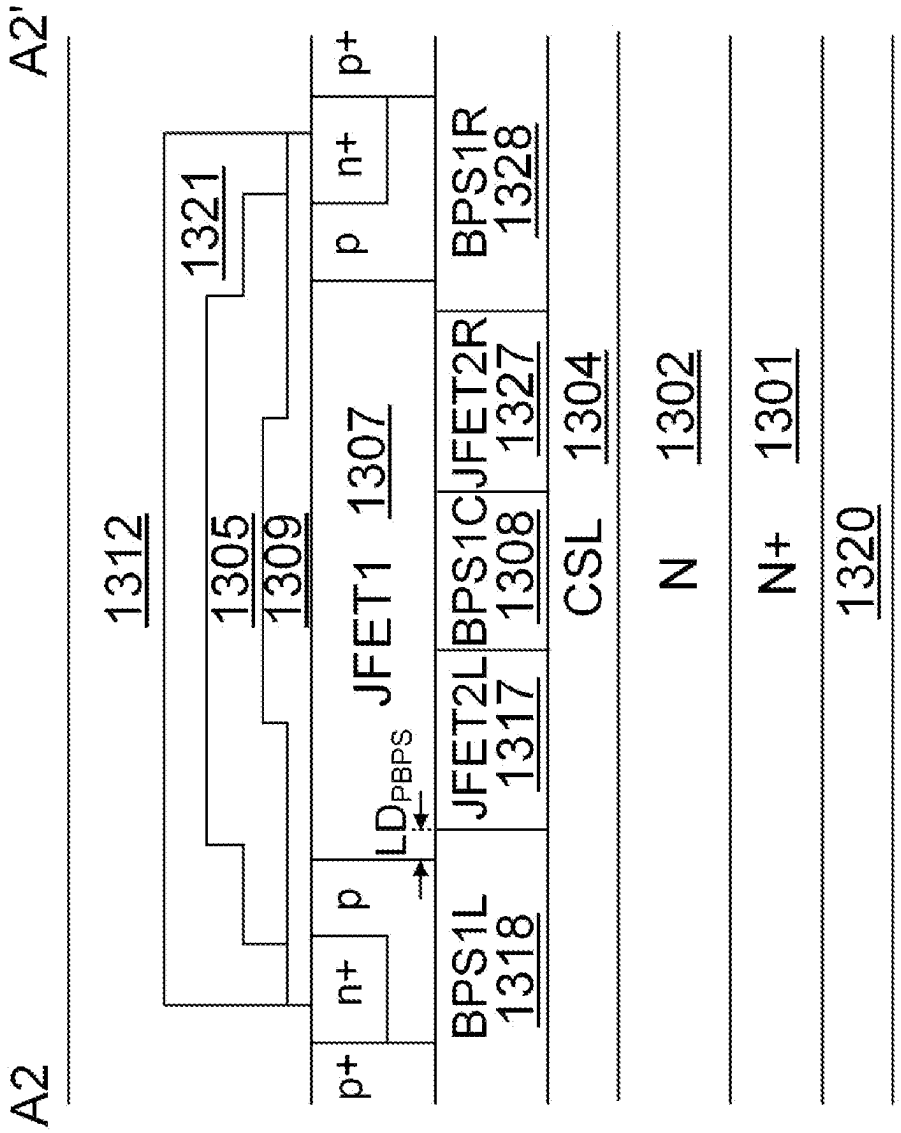

A2
A2'
1312
1321
1305
1309
p+
n+
p
LD$_{PBPS}$
JFET1 1307
p
n+
p+
BPS1L
1318
JFET2L
1317
BPS1C
1308
JFET2R
1327
BPS1R
1328
CSL 1304
N 1302
N+ 1301
1320

Fig. 3G

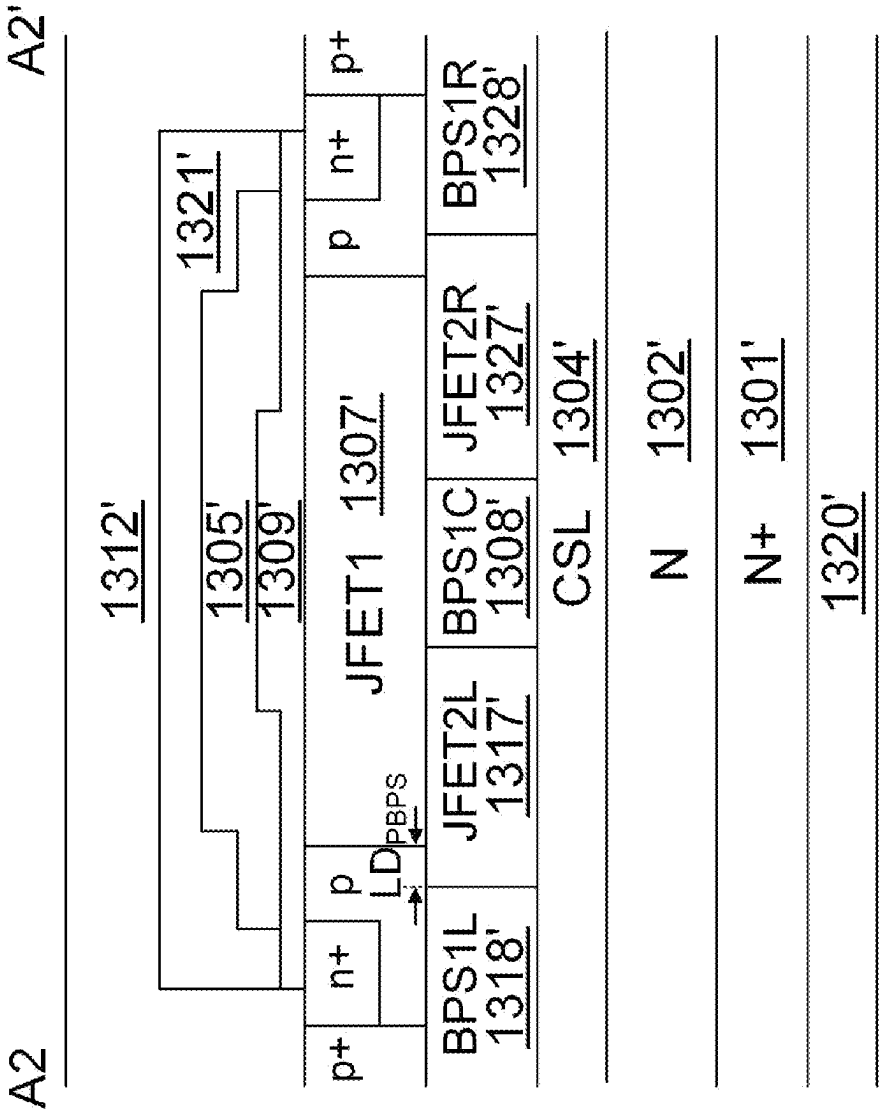

A2
A2'
1312'
1321'
1305'
1309'
p+
n+
p
LDPBPS
JFET1 1307'
p
n+
p+
BPS1L 1318'
JFET2L 1317'
BPS1C 1308'
JFET2R 1327'
BPS1R 1328'
CSL 1304'
N 1302'
N+ 1301'
1320'

Fig. 3H

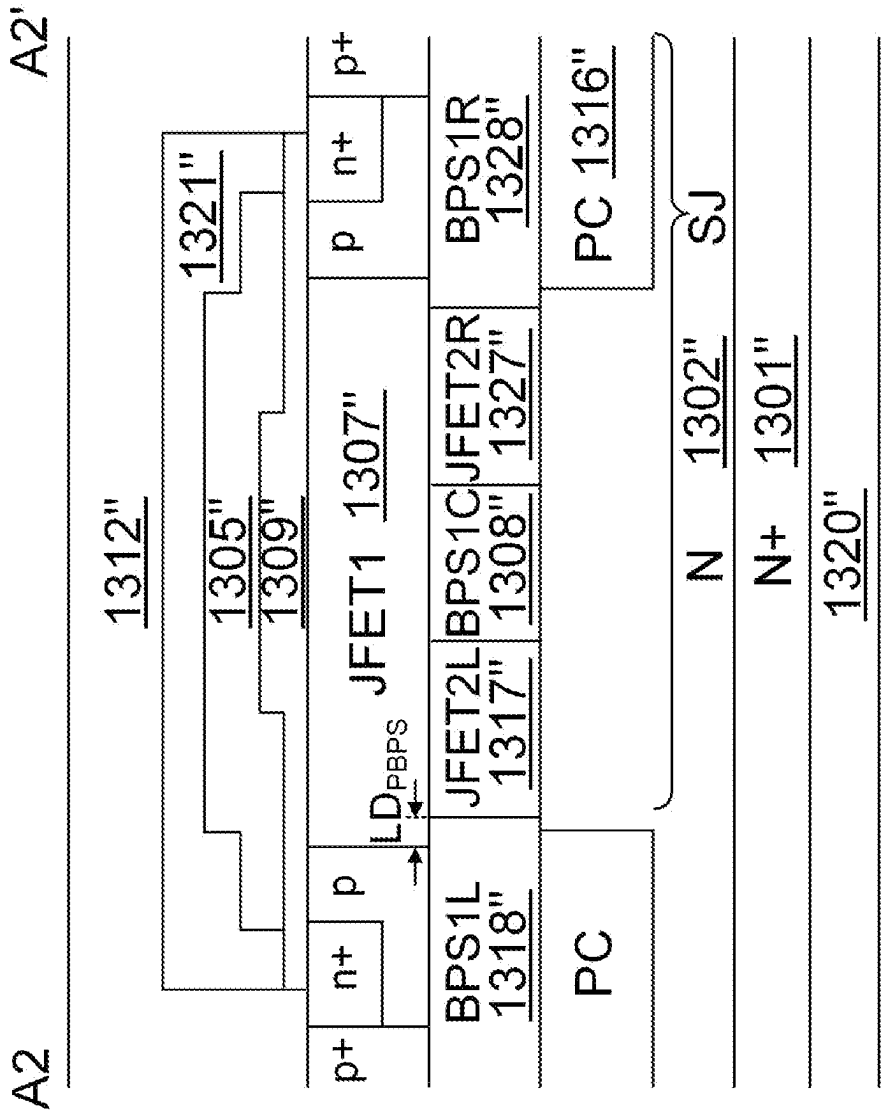

A2
A2'
1312"
1321"
1305"
1309"
p+
n+
p
LD$_{PBPS}$
JFET1 1307"
p
n+
p+
BPS1L 1318"
JFET2L 1317"
BPS1C 1308"
JFET2R 1327"
BPS1R 1328"
PC
PC 1316"
SJ
N 1302"
N+ 1301"
1320"

Fig. 3I

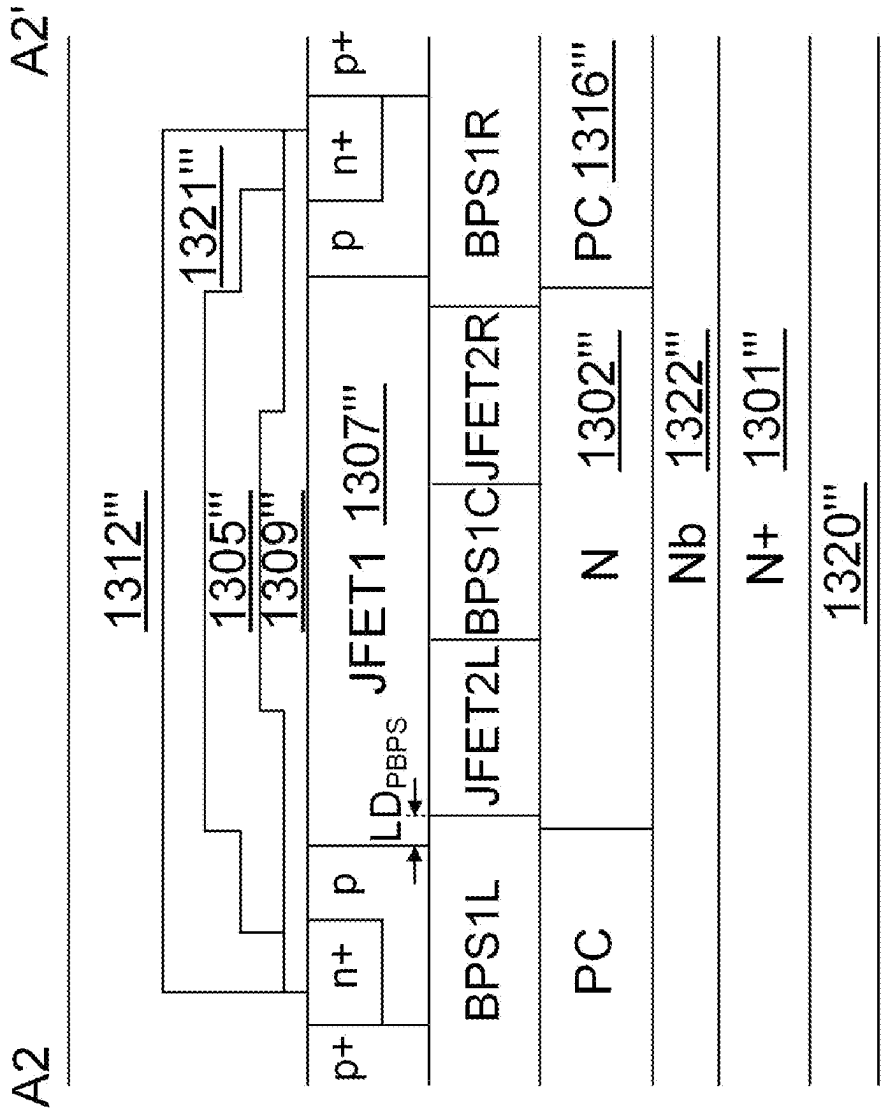

A2
A2'
1312'''
1305'''
1309'''
1321'''
p+
n+
p
JFET1 1307'''
LD_PBPS
p
n+
p+
BPS1L
JFET2L
BPS1C
JFET2R
BPS1R
PC
N 1302'''
PC 1316'''
Nb 1322'''
N+ 1301'''
1320'''

Fig. 3J

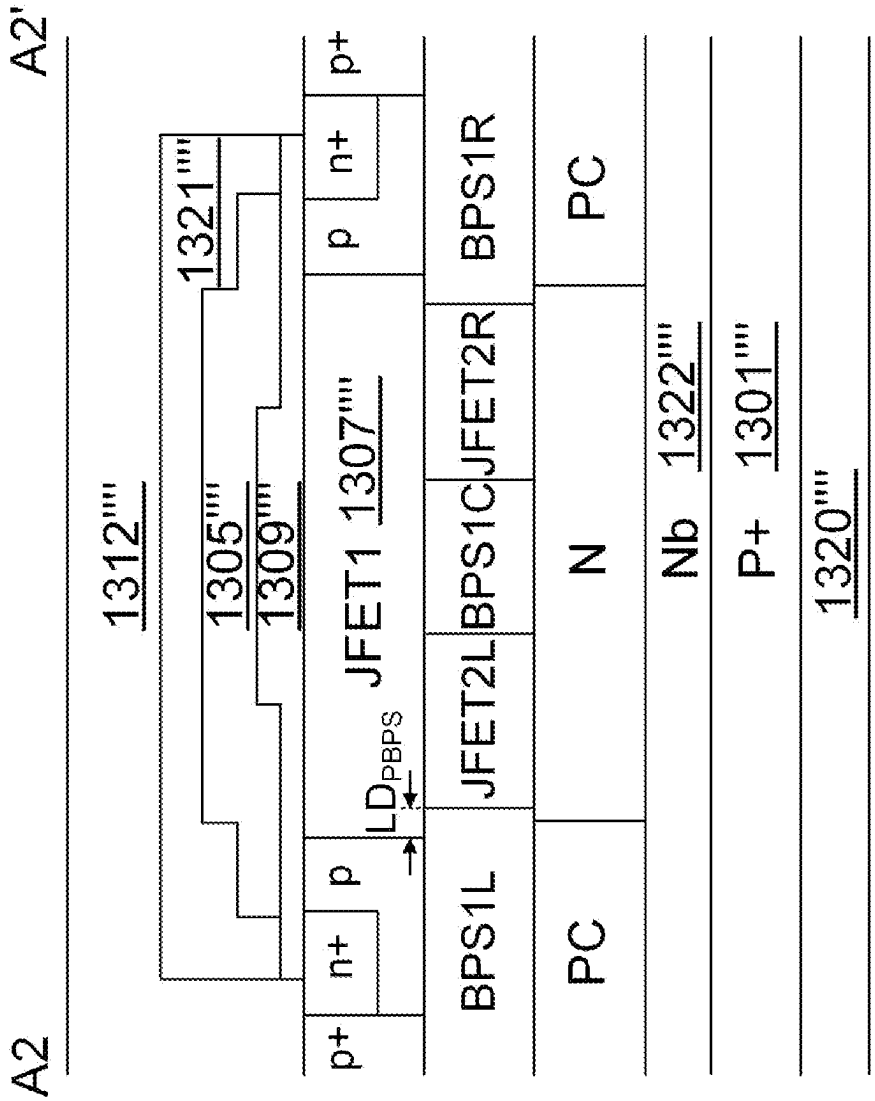

A2
A2'
1312''''
1321''''
1305''''
1309''''
p+
n+
p
LD$_{PBPS}$
JFET1 1307''''
p
n+
p+
BPS1L
JFET2L
BPS1C
JFET2R
BPS1R
PC
N
PC
Nb 1322''''
P+ 1301''''
1320''''

Fig. 3K

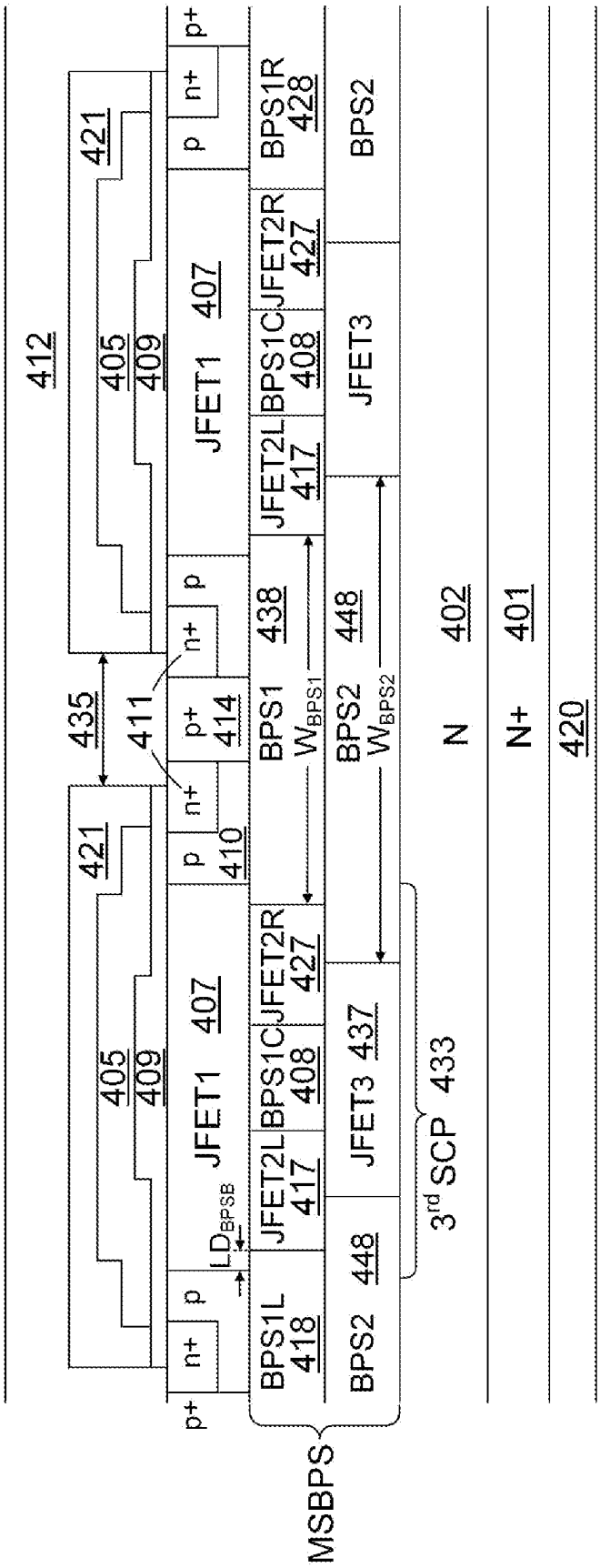

412
421
405
409
435
411
p+
n+
p
JFET1 407
JFET2L 417
BPS1C 408
JFET2R 427
BPS1R 428
BPS1 438
BPS2 448
JFET3
JFET3 437
BPS1L 418
BPS2 448
LD_{BPSB}
W_{BPS1}
W_{BPS2}
3rd SCP 433
MSBPS
N 402
N+ 401
420

Fig. 4A

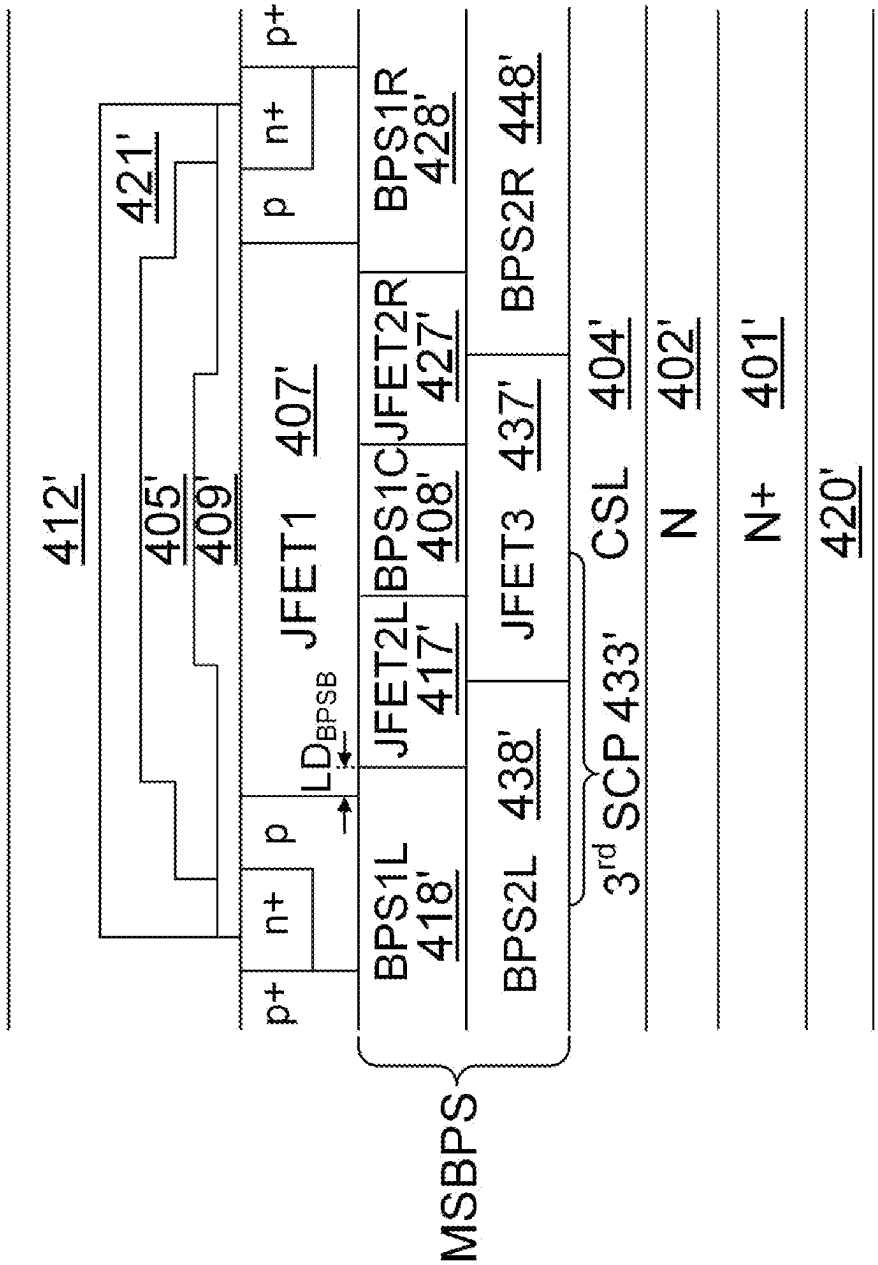

412'
421'
405'
409'
p+
n+
p
JFET1 407'
p
n+
p+
LD_BPSB
BPS1L 418'
JFET2L 417'
BPS1C 408'
JFET2R 427'
BPS1R 428'
BPS2L 438'
JFET3 437'
BPS2R 448'
3rd SCP 433'
CSL 404'
N 402'
N+ 401'
420'
MSBPS

Fig. 4B

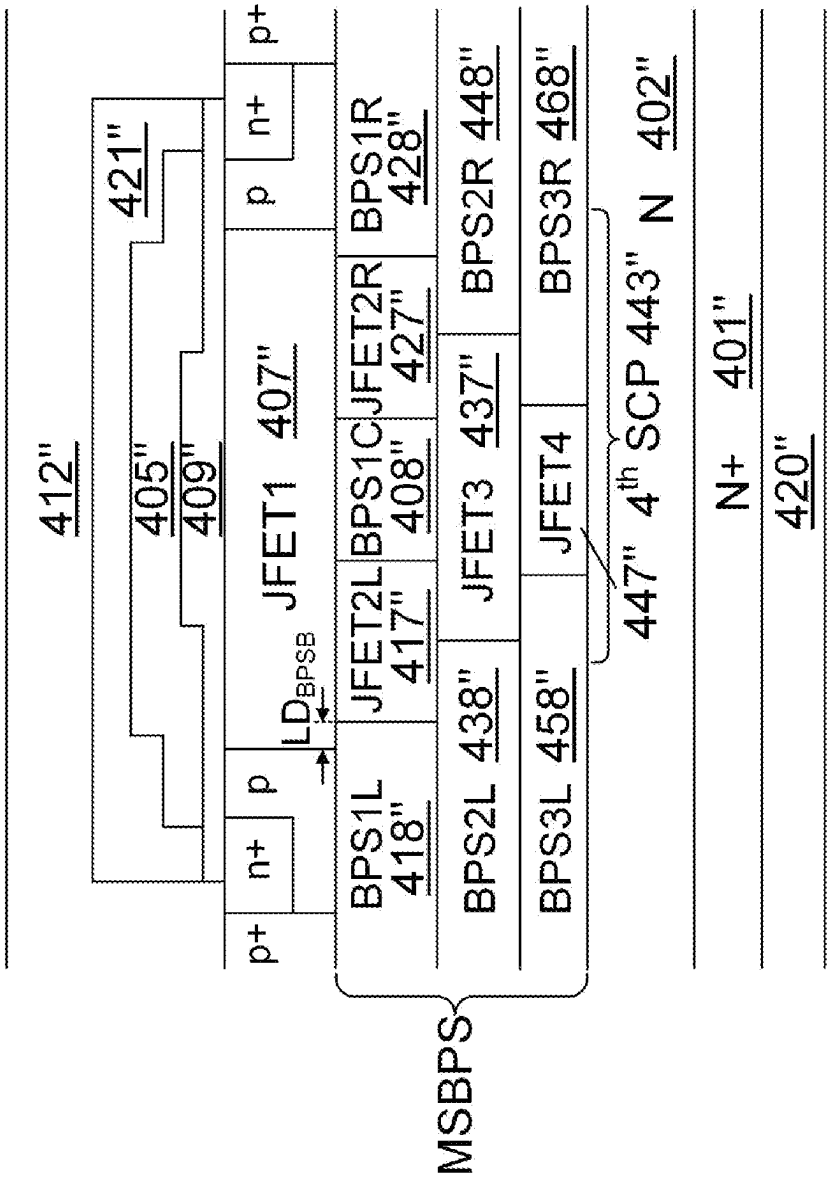

412"
421"
405"
409"
p+
n+
p
p
n+
p+
JFET1 407"
LD_BPSB
BPS1L 418"
JFET2L 417"
BPS1C 408"
JFET2R 427"
BPS1R 428"
BPS2L 438"
JFET3 437"
BPS2R 448"
BPS3L 458"
JFET4
BPS3R 468"
447" 4th SCP 443"
N 402"
N+ 401"
420"
MSBPS

Fig. 4C

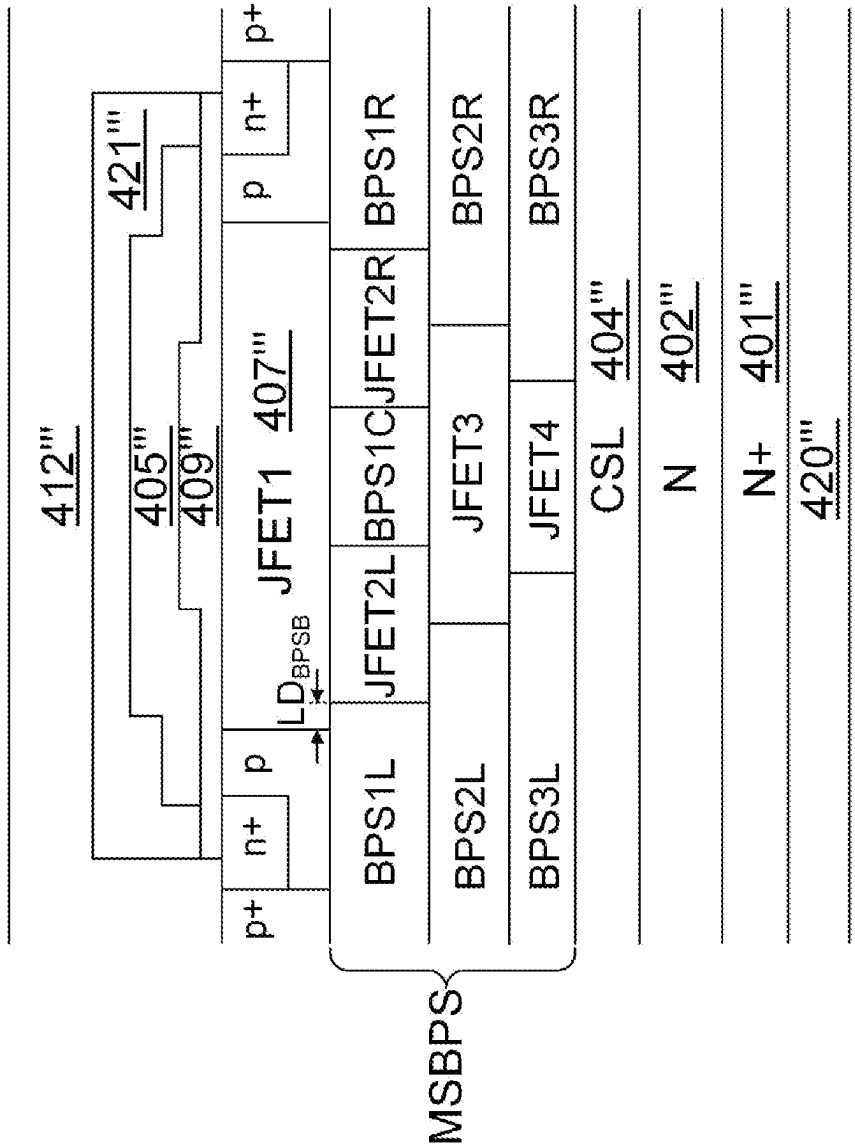

412'''
421'''
405'''
409'''
p+
n+
p
JFET1
407'''
p
n+
p+
LD_BPSB
BPS1L
JFET2L
BPS1C
JFET2R
BPS1R
BPS2L
JFET3
BPS2R
BPS3L
JFET4
BPS3R
CSL
404'''
N
402'''
N+
401'''
420'''
MSBPS

Fig. 4D

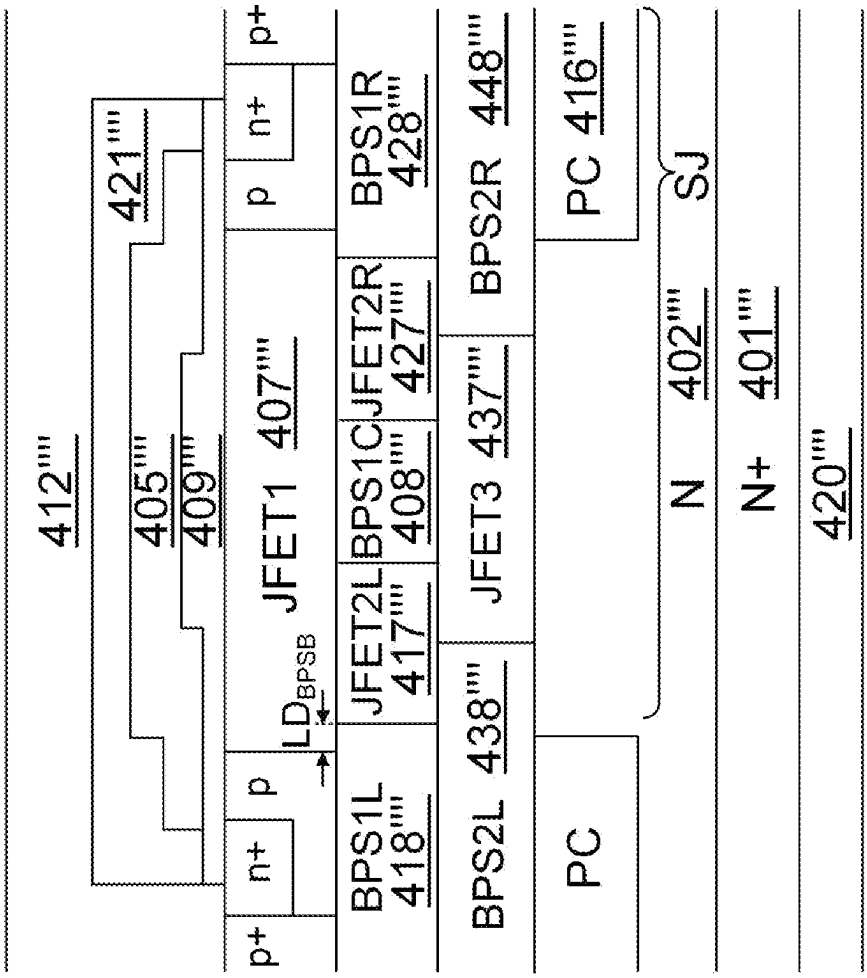

412''''
421''''
405''''
409''''
p+
n+
p
LD$_{BPSB}$
JFET1 407''''
p
n+
p+
BPS1L 418''''
JFET2L 417''''
BPS1C 408''''
JFET2R 427''''
BPS1R 428''''
BPS2L 438''''
JFET3 437''''
BPS2R 448''''
PC
PC 416''''
N 402''''
SJ
N+ 401''''
420''''

Fig. 4E

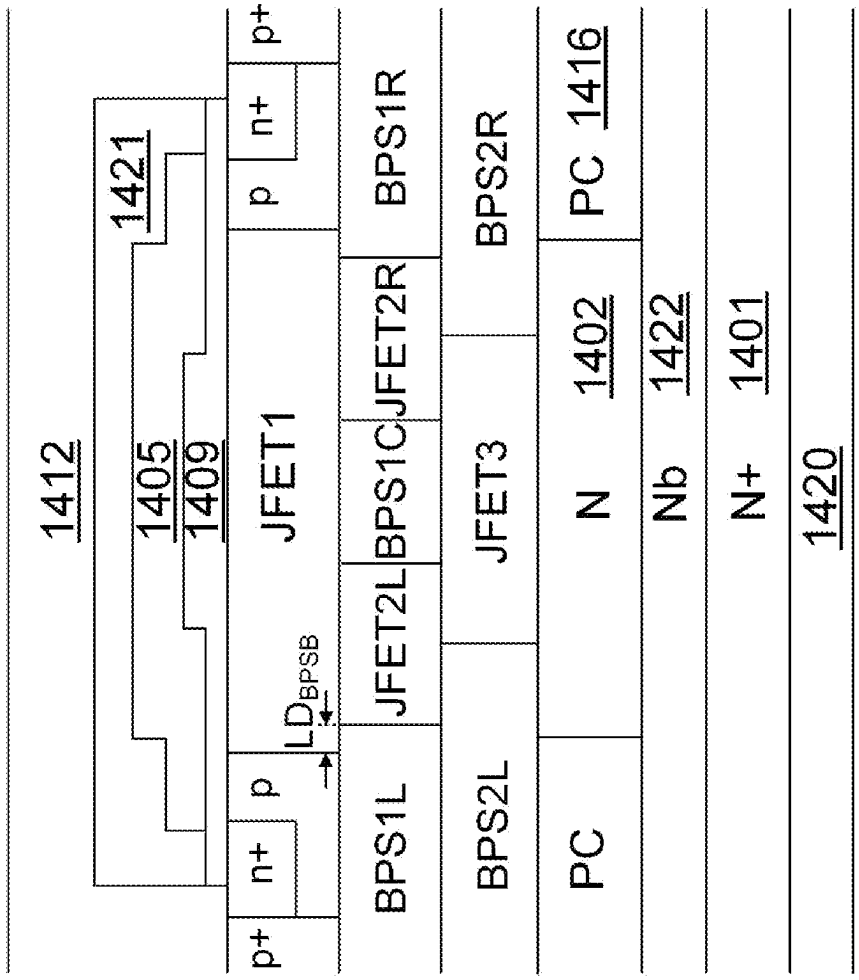

1412
1421
1405
1409
p+
n+
p
JFET1
p
n+
p+
LD_BPSB
BPS1L
JFET2L
BPS1C
JFET2R
BPS1R
BPS2L
JFET3
BPS2R
PC
N
1402
PC
1416
Nb
1422
N+
1401
1420

Fig. 4F

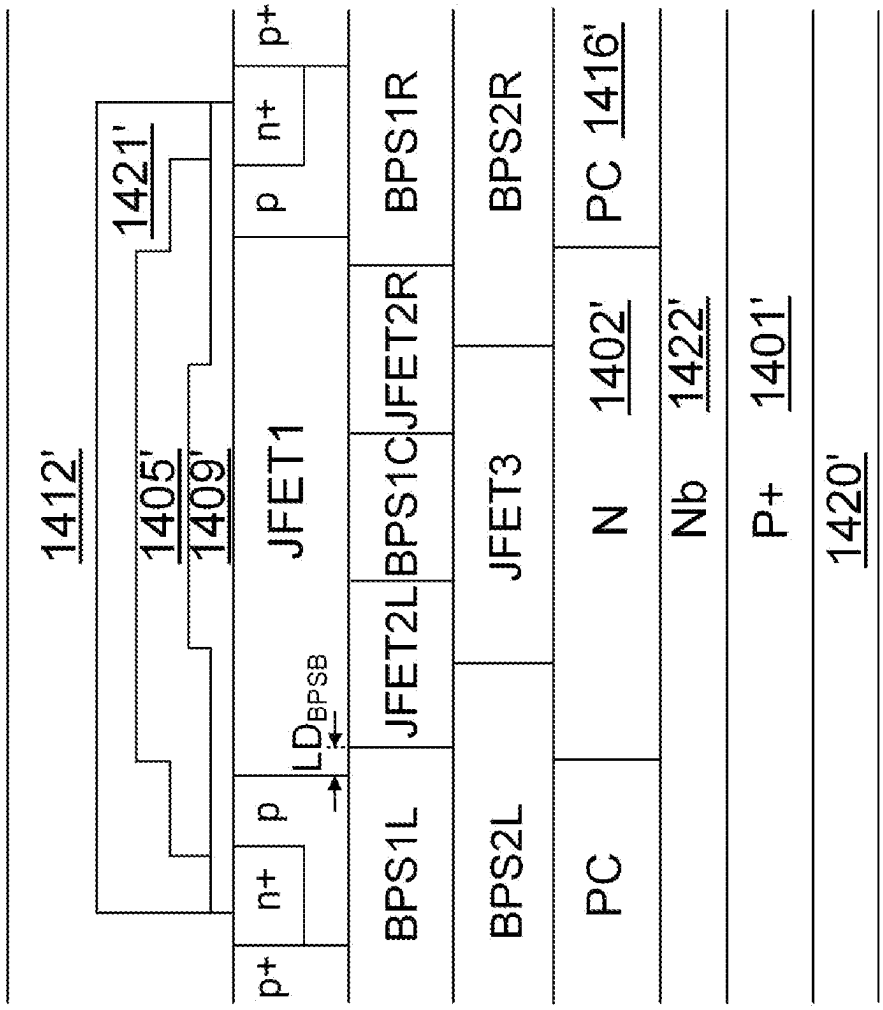

1412'
1405'
1409'
1421'
p+
n+
p
JFET1
p
n+
p+
LD_BPSB
BPS1L
JFET2L
BPS1C
JFET2R
BPS1R
BPS2L
JFET3
BPS2R
PC
N
1402'
PC
1416'
Nb
1422'
P+
1401'
1420'

Fig. 4G

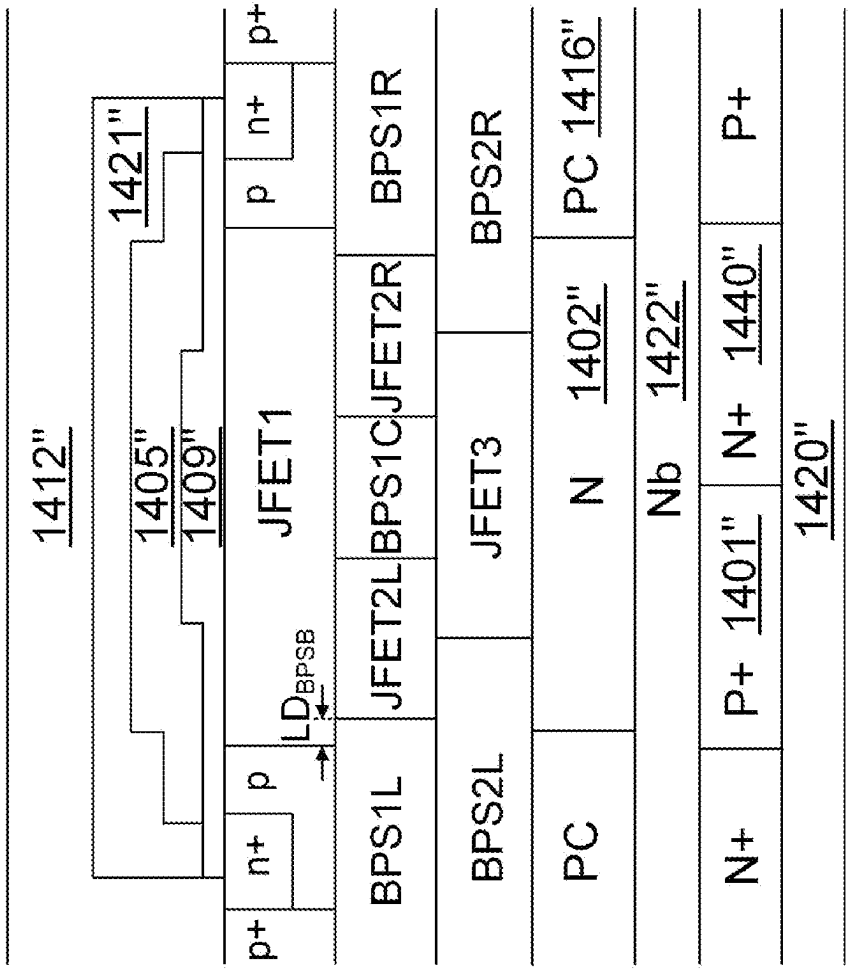

1412"
1421"
1405"
1409"
p+
n+
p
JFET1
p
n+
p+
LD_BPSB
BPS1L
JFET2L
BPS1C
JFET2R
BPS1R
BPS2L
JFET3
BPS2R
PC
N
1402"
PC 1416"
Nb
1422"
N+
P+
1401"
N+
1440"
P+
1420"

Fig. 4H

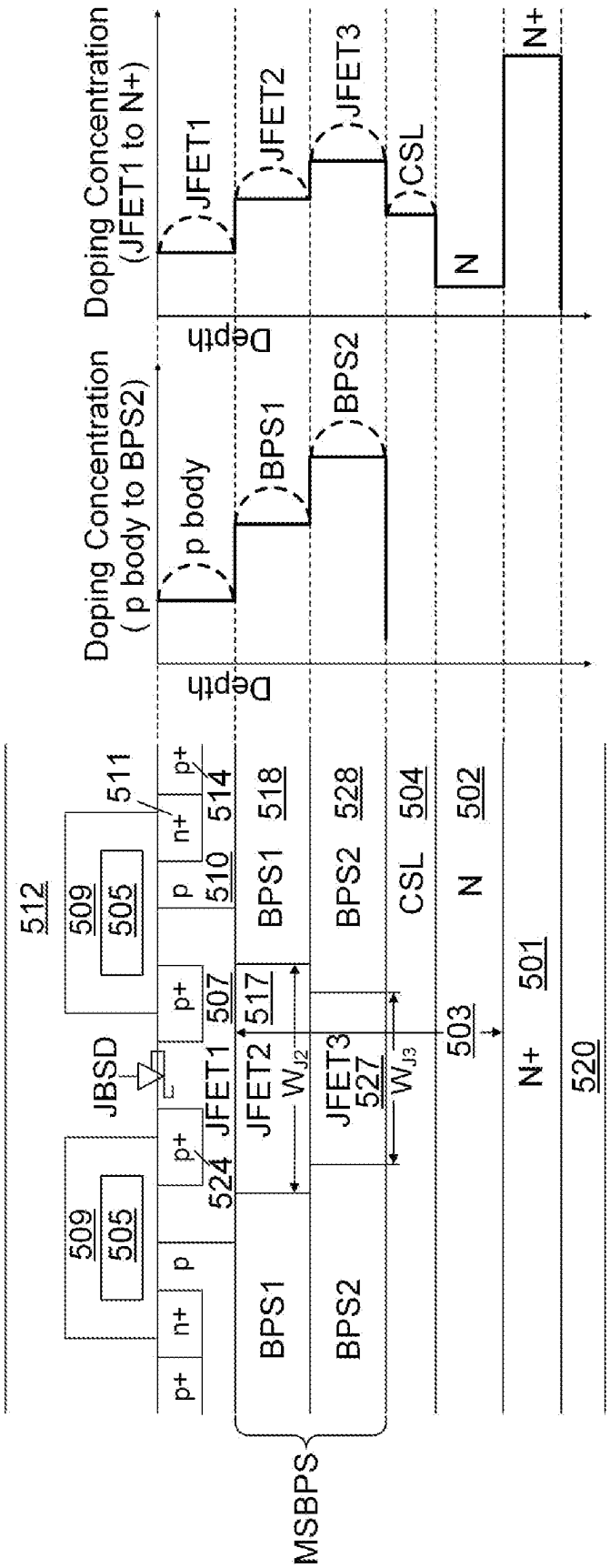

Doping Concentration
(JFET1 to N+)
Doping Concentration
(p body to BPS2)
Depth
Depth
JFET1
JFET2
JFET3
CSL
N
N+
p body
BPS1
BPS2
512
509
505
511
p+
n+
p
510
514
518
528
504
502
507
517
527
503
501
520
JBSD
524
JFET1
JFET2
JFET3
W_{J2}
W_{J3}
MSBPS

Fig. 5A

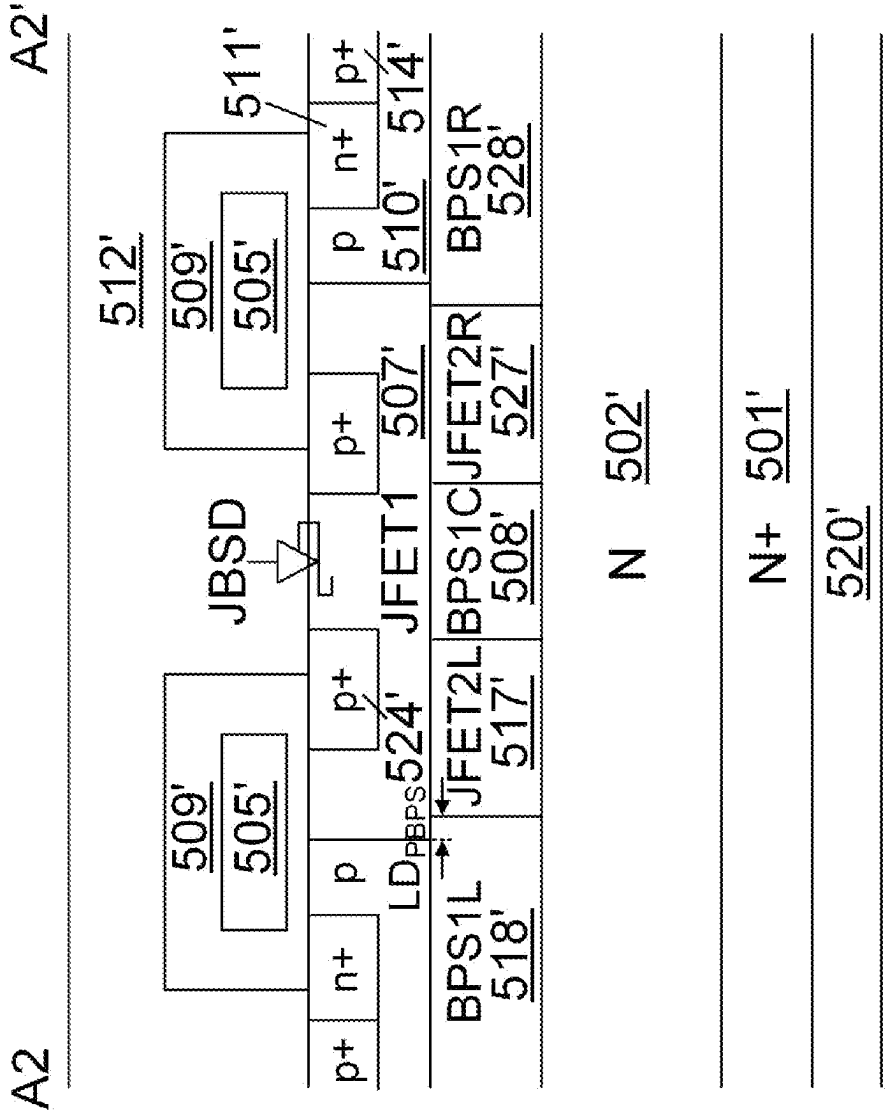

A2
A2'
512'
509'
505'
509'
505'
511'
p+
n+
p
p+
p+
p
n+
p+
JBSD
JFET1
507'
510'
514'
524'
LD
PBPS
BPS1L
518'
JFET2L
517'
BPS1C
508'
JFET2R
527'
BPS1R
528'
N
502'
N+
501'
520'

Fig. 5B

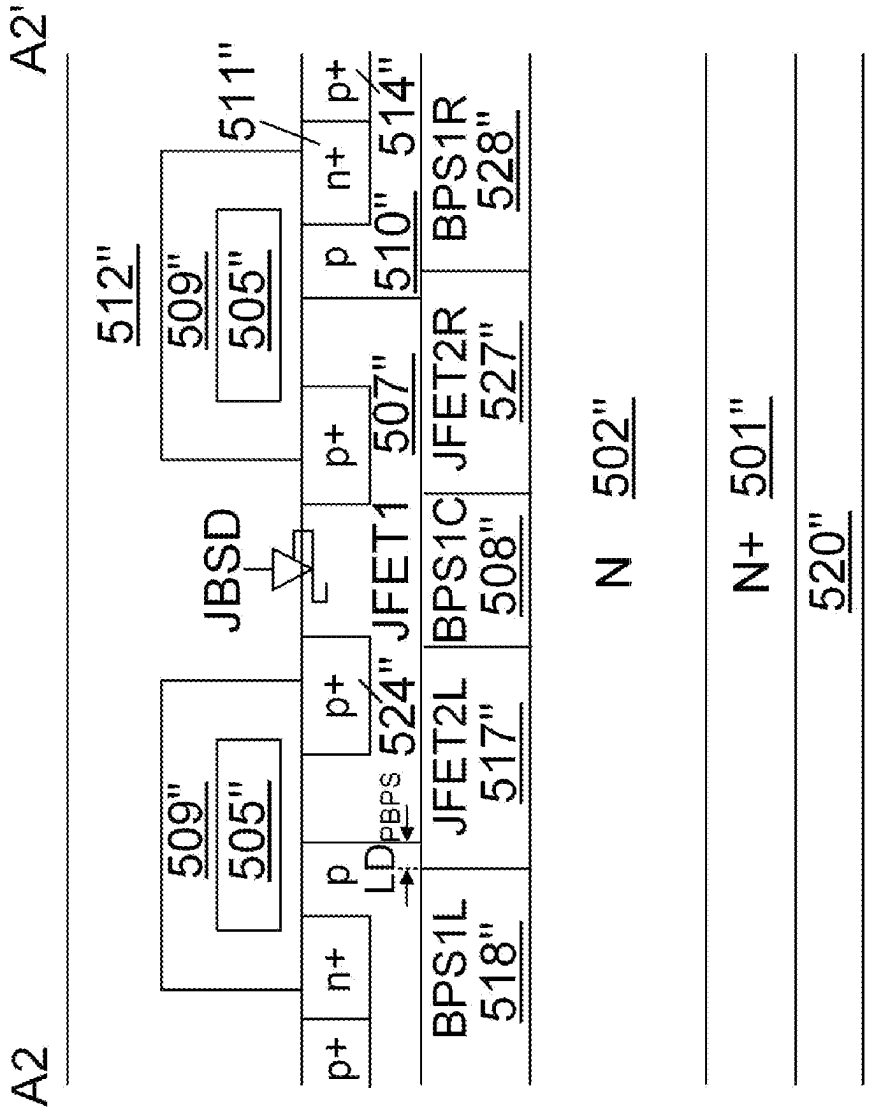

A2
A2'
512"
509"
505"
509"
505"
511"
JBSD
p+
n+
p
LD
PBPS
p+
524"
JFET1
p+
507"
p
510"
n+
p+
514"
BPS1L
518"
JFET2L
517"
BPS1C
508"
JFET2R
527"
BPS1R
528"
N 502"
N+ 501"
520"

Fig. 5C

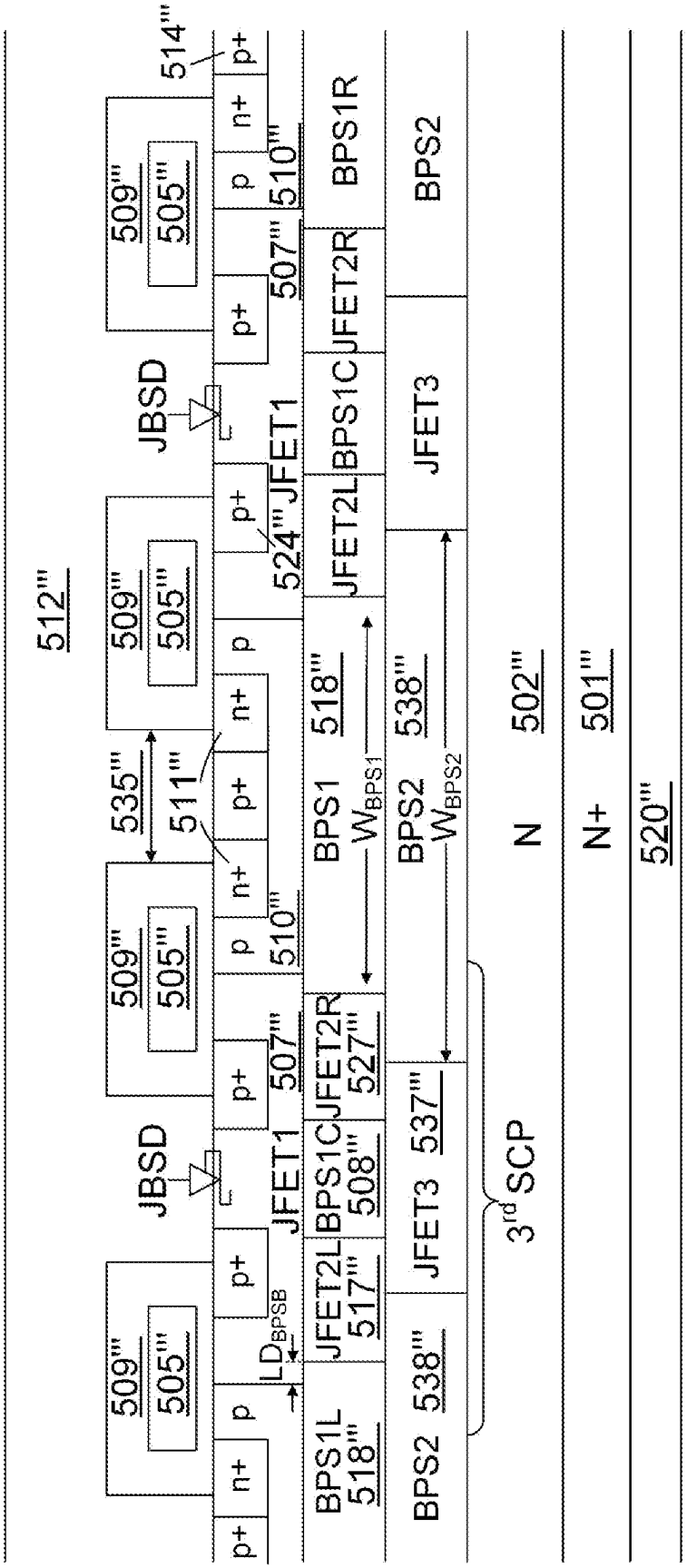

512'''
514'''
p+
n+
p
509'''
505'''
JBSD
JFET1
507'''
510'''
524'''
JFET2L
BPS1C
JFET2R
BPS1R
BPS2
JFET3
535'''
511'''
518'''
538'''
BPS1
BPS2
W_{BPS1}
W_{BPS2}
527'''
537'''
508'''
517'''
BPS1L
LD_{BPSB}
3rd SCP
N
502'''
N+
501'''
520'''

Fig. 5D

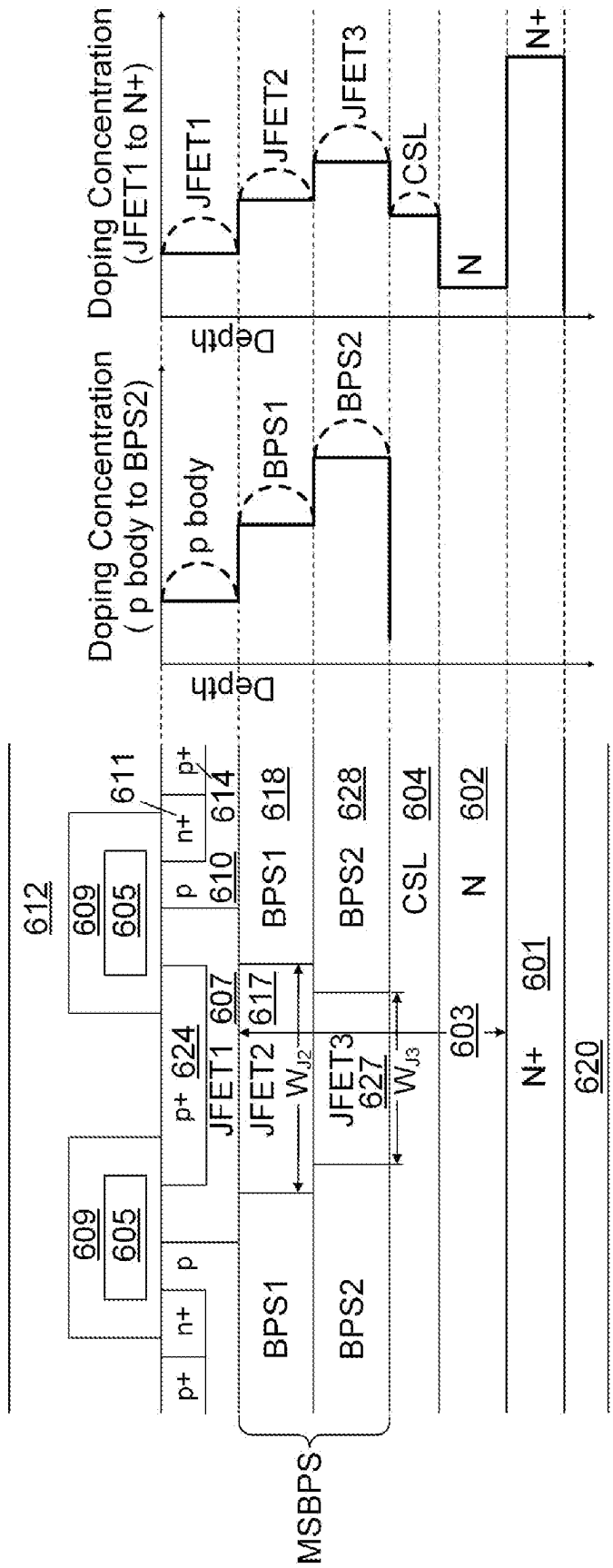

Doping Concentration
(p body to BPS2)
Doping Concentration
(JFET1 to N+)
Depth
p body
BPS1
BPS2
JFET1
JFET2
JFET3
CSL
N
N+
p+
n+
p
612
609
605
611
614
610
618
628
604
602
607
617
624
627
603
601
620
W_{J2}
W_{J3}
MSBPS

Fig. 6A

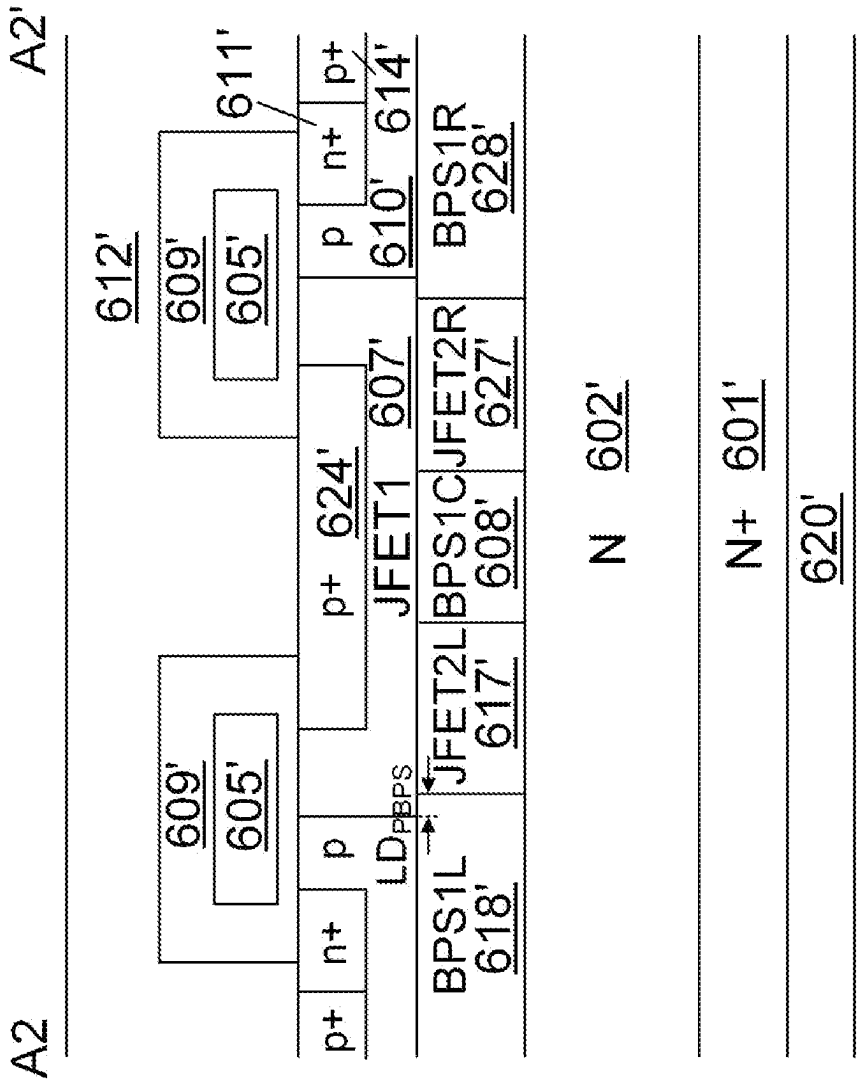

A2
A2'
612'
609'
605'
609'
605'
611'
p+
n+
p
p+
624'
JFET1
607'
p
n+
p+
610'
614'
BPS1L
618'
JFET2L
617'
BPS1C
608'
JFET2R
627'
BPS1R
628'
LDPBPS
N 602'
N+ 601'
620'

Fig. 6B

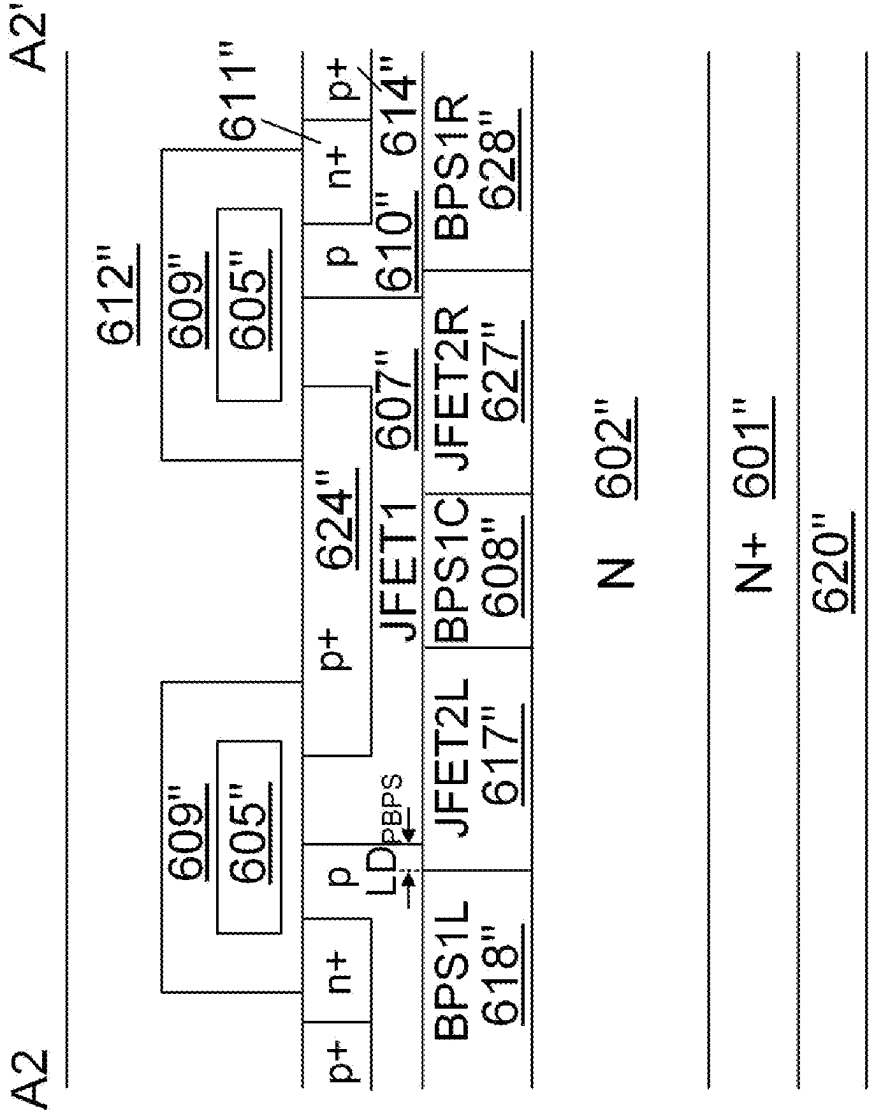

A2
A2'
612"
609"
605"
609"
605"
611"
p+
n+
p
LD
PBPS
p+
624"
JFET1
607"
p
610"
n+
p+
614"
BPS1L
618"
JFET2L
617"
BPS1C
608"
JFET2R
627"
BPS1R
628"
N
602"
N+
601"
620"

Fig. 6C

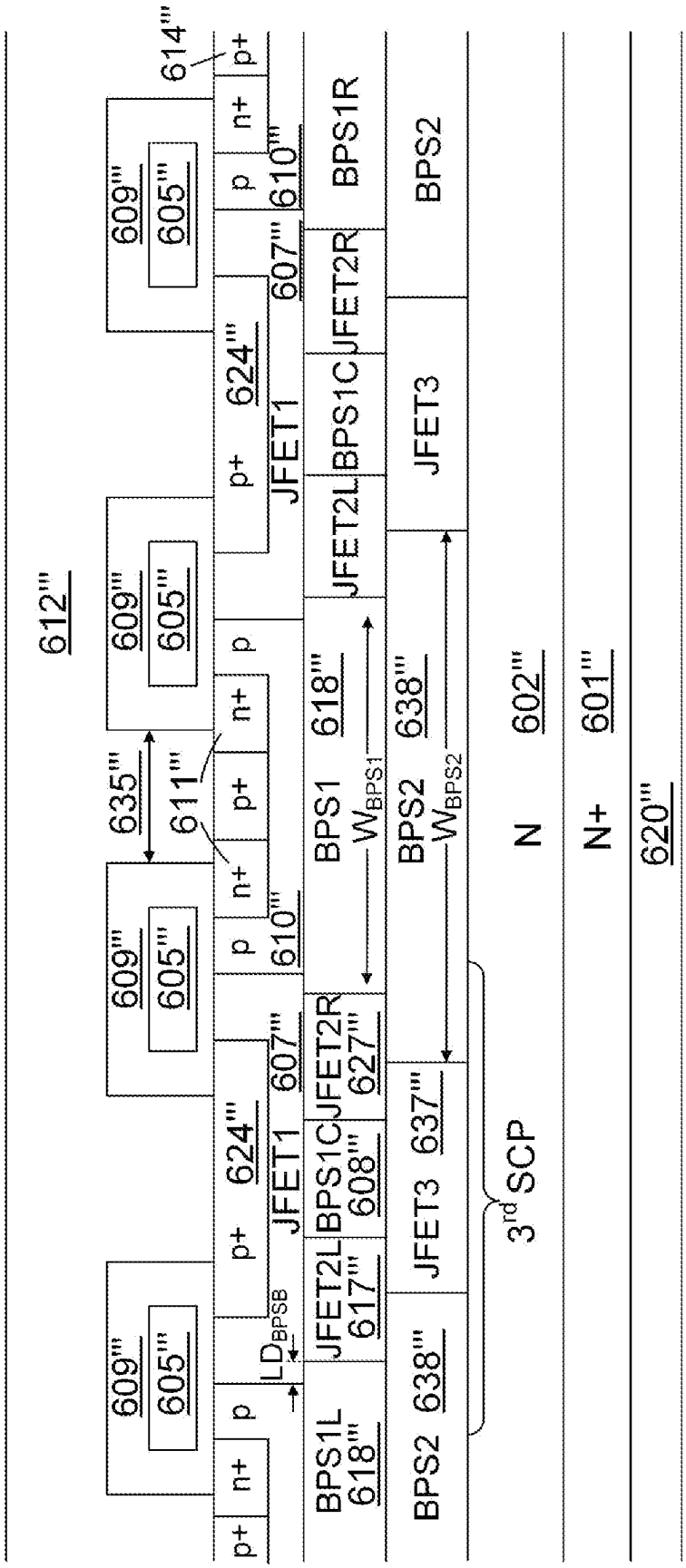

612'''
614'''
609'''
605'''
635'''
611'''
624'''
607'''
610'''
618'''
638'''
627'''
608'''
617'''
637'''
602'''
601'''
620'''
p+
n+
p
JFET1
JFET2L
JFET2R
BPS1C
BPS1R
JFET3
BPS1
BPS2
BPS1L
N
N+
W_{BPS1}
W_{BPS2}
LD_{BPSB}
3rd SCP

Fig. 6D

SIC PLANAR MOSFETS WITH IMPROVED PERFORMANCE STRUCTURES

FIELD OF THE INVENTION

[0001] This invention relates generally to semiconductor devices having multiple stepped buried P-shield (MSBPS) regions, and more particularly, to a silicon carbide (SiC) planar Metal Oxide Semiconductor Field Effect Transistor (MOSFET) having a first buried P-shield (BPS) region including one center portion and two side portions for the gate oxide electric field and saturation current reductions. Moreover, a Junction Barrier Schottky diode (JBSD) is integrated with the SiC MOSFET in a location between two adjacent split gate electrodes for the reverse conduction switching loss reduction.

BACKGROUND OF THE INVENTION

[0002] Because of the physical properties of SiC, SiC-MOSFETs can achieve a higher breakdown voltage, a lower on-resistance and a higher switching speed than Si-MOSFETs. However, SiC-MOSFETs have a much higher electric-field strength at the gate oxide than Si-MOSFETs because the breakdown electric field strength of SiC is 10 times higher than that of Si. To resolve this problem, a P+ shielding region is disposed under an N+ source region as well as a P-base region as shown in FIG. **1** (FIG. **10.1** of Silicon Carbide Power Devices, B Jayant Baliga's book, page 260) as a field relaxing layer near the gate oxide. Unfortunately, a short-circuit (SC) failure is still the most serious reliable issue for the shielded SiC-MOSFETs device due to the much smaller device area comparing with Si-MOSFETs resulting in a higher junction temperature before the failure. During the SC transient process, the high direct current (DC) voltage and the large saturation current would be applied to the device, resulting in the device's performance degradation or even severe burnout failure if the devices do not have a sufficient SC withstand capability.

[0003] Therefore, there is still a need in the art of the SiC-MOSFETs design and fabrication to provide a novel cell structure, device configuration and manufacturing process that would make the SiC-MOSFETs have a lower electric-field strength at the gate oxide, achieve a lower specific on-resistance and have a lower saturation current for the short circuit capability improvement.

SUMMARY OF THE INVENTION

[0004] The present invention discloses a SiC planar MOSFET having multiple stepped buried P-shield (MSBPS) regions of a second conductivity adjoining a body region with different doping concentrations increasing stepwise in a direction from the body region toward a substrate, and multiple stepped Field Effect Transistor (JFET) regions of a first conductivity type sandwiched between the adjacent MSBPS regions with different widths decreasing stepwise in a direction from the body region toward the substrate; multiple saturation current pinching (SCP) structures formed in the multiple stepped JFET (MSJ) regions between the adjacent MSBPS regions clamp the potential, thus limiting the saturation current in a forward conduction stage for the short-circuit capability improvement. The gate oxide electric field strength is also reduced by the existence of the MSBPS regions as a high electric field is shifted from a channel region to a central gate electrode region.

[0005] According to one aspect, the invention features a SiC device comprising a plurality of unit cells with each unit cell in an active area, comprising: an epitaxial layer grown on a substrate; a source region of a first conductivity type formed at a top portion of the epitaxial layer and encompassed in a body region of a second conductivity type; a planar gate electrode made of a doped poly-silicon layer padded by a gate oxide; a first type JFET (JFET1) region of the first conductivity type formed between the two adjacent body regions; MSBPS regions of the second conductivity type adjoining the body region with different doping concentrations increasing stepwise in a direction from the body region toward the substrate; The MSJ regions of the first conductivity type sandwiched between the two adjacent MSBPS regions with different widths decreasing stepwise in a direction from the body region toward the substrate; and an N-drift (ND) region of the first conductivity type formed between the substrate and the JFET1 region.

[0006] According to another aspect, in some preferred embodiments, the MSJ regions have different doping concentrations increasing stepwise in a direction from the JFET1 region toward the substrate.

[0007] According to another aspect, in some preferred embodiments, the JFET1 region has a width greater than the widths of the MSJ regions, and has a doping concentration less than doping concentrations of the MSJ regions.

[0008] According to another aspect, in some preferred embodiments, the MSBPS regions comprising at least two BPS regions and the MSJ regions comprising at least two JFET regions.

[0009] According to another aspect, the invention also features a SiC device comprising a plurality of unit cells with each unit cell in an active area, comprising: an epitaxial layer grown on a substrate; a source region of a first conductivity type formed at a top portion of the epitaxial layer and encompassed in a body region of a second conductivity type; a planar gate electrode made of a doped poly-silicon layer padded by a gate oxide; a first type JFET (JFET1) region of the first conductivity type formed between the two adjacent body regions; a first BPS (BPS1) region of the second conductivity type formed below the body region and the JFET1 region having a left side portion BPS1L, a center portion BPS1C, a right side portion BPS1R and connection portions BPS1G; the BPS1G connecting the BPS1C together with the BPS1L and/or the BPS1R to a source metal through the body region; a second type JFET (JFET2) region having a left side portion JFET2L formed between the BPS1L and the BPS1C, and a right side portion JFET2R formed between the BPS1C and the BPS1R; and an ND region of the first conductivity type formed between the substrate and the JFET1 region.

[0010] According to another aspect, the invention features an improved SiC device having a single gate electrode padded with a gate oxide having a central thick gate oxide under a center of the single gate electrode.

[0011] According to another aspect, the invention features an improved SiC device with a planar gate electrode having two split gate electrodes in each unit cell, and further comprising a Junction Barrier Schottky diode (JBSD) embedded between the two adjacent gate electrodes by making a Schottky contact with the JFET1 region and the two P+ regions of a second conductivity type within the JFET1 region.

[0012] According to another aspect, the invention features an improved SiC device with a planar gate electrode having two split gate electrodes in each unit cell, and further comprising a P/N (P-type/N-type) junction diode embedded between the two adjacent split gate electrodes by making a p+ region of a second conductivity type within the JFET1 region of a first conductivity type, wherein the p+ region is spaced apart from the body region for the gate oxide electric field reduction.

[0013] According to another aspect, the present invention also features a SiC device further comprising a second BPS (BPS2) region of the second conductivity type formed below the BPS1 region and a third type JFET (JFET3) region formed between the two adjacent BPS2 regions; the BPS2 region having a doping concentration higher than that of the BPS1 region, and having a width greater than that of the BPS1 region.

[0014] According to another aspect, the present invention also features a SiC device further comprising a CSL below the BPS2 and the JFET3 regions with a doping concentration higher than that of the ND region.

[0015] According to another aspect, in some preferred embodiments, the JFET2 region has a doping concentration higher than that of the JFET1 region.

[0016] According to another aspect, the present invention also features a SiC device further comprising a super junction structure comprising a P column region of the second conductivity type disposed above the substrate.

[0017] According to another aspect, in some preferred embodiments, the substrate has the first conductivity type and further comprises a buffer layer of the first conductivity type with a resistivity Rb sandwiched between the substrate and the epitaxial layer with a resistivity R, wherein R<Rb.

[0018] According to another aspect, in some preferred embodiments, the substrate has the second conductivity type and further comprises a buffer layer of the first conductivity type with a resistivity Rb sandwiched between the substrate and the epitaxial layer with a resistivity R, wherein R>Rb.

[0019] According to another aspect, in some preferred embodiments, the substrate has the second conductivity type, further comprises a buffer layer of the first conductivity type formed sandwiched between the substrate and the epitaxial layer; a plurality of heavily doped regions of the first conductivity type in the substrate to form a plurality of alternating P+ and N+ regions in the substrate.

[0020] These and other objects and advantages of the present invention will no doubt become obvious to those of ordinary skill in the art after having read the following detailed description of the preferred embodiment, which is illustrated in the various drawing figures.

BRIEF DESCRIPTION OF THE DRAWINGS

[0021] The accompanying drawings, which are included to provide a further understanding of the invention and are incorporated in and constitute a part of this specification, illustrate exemplary embodiments of the invention and together with the description serve to explain the principles of the invention. In the drawings:

[0022] FIG. **1** is a cross-sectional view of a SiC shielded MOSFET device of a prior art.

[0023] FIG. **2**A is a cross-sectional view of a preferred embodiment wherein doping concentration variations of the P type regions (from p body region to BPS2 region) and the N type regions (from JFET1 region to N+ substrate) are depicted separately along the vertical direction according to the present invention.

[0024] FIG. **2**B is a cross-sectional view of another preferred embodiment with a current spreading layer (CSL) wherein doping concentration variations of the P type regions (from p body region to BPS2 region) and the N type regions (from JFET1 region to N+ substrate) are depicted separately along the vertical direction according to the present invention.

[0025] FIG. **2**C is a cross-sectional view of another preferred embodiment with three MSBPS regions wherein doping concentration variations of the P type regions (from p body region to BPS2 region) and the N type regions (from JFET1 region to N+ substrate) are depicted separately along the vertical direction according to the present invention.

[0026] FIG. **2**D is a cross-sectional view of another preferred embodiment having a super junction structure in FIG. **2**A according to the present invention.

[0027] FIG. **2**E is a cross-sectional view of another preferred embodiment having a buffer layer in FIG. **2**D according to the present invention.

[0028] FIG. **2**F is a cross-sectional view of another preferred embodiment according to the present invention, which is same as FIG. **2**E except that the substrate conductivity type is P type.

[0029] FIG. **2**G is a cross-sectional view of another preferred embodiment according to the present invention, which is same as FIG. **2**F except that the substrate has a plurality of alternating P+ and N+ regions.

[0030] FIG. **3**A is a top view of a preferred embodiment according to the present invention.

[0031] FIG. **3**B is a cross-sectional view of a SiC device showing a preferred embodiment along A**1**-A**1'** line of FIG. **3**A according to the present invention.

[0032] FIG. **3**C is a cross-sectional view of a SiC device showing a preferred embodiment along B**1**-B**1'** line of FIG. **3**A according to the present invention.

[0033] FIG. **3**D is a top view of another preferred embodiment according to the present invention.

[0034] FIG. **3**E is a cross-sectional view of a SiC device showing a preferred embodiment along A**2**-A**2'** line of FIG. **3**D according to the present invention.

[0035] FIG. **3**F is a cross-sectional view of a SiC device showing a preferred embodiment along B**2**-B**2'** line of FIG. **3**D according to the present invention.

[0036] FIG. **3**G is a cross-sectional view of a SiC device showing another preferred embodiment along A**2**-A**2'** line of FIG. **3**D according to the present invention.

[0037] FIG. **3**H is a cross-sectional view of a SiC device showing another preferred embodiment along A**2**-A**2'** line of FIG. **3**D according to the present invention.

[0038] FIG. **3**I is a cross-sectional view of a SiC device showing another preferred embodiment along A**2**-A**2'** line of FIG. **3**D according to the present invention.

[0039] FIG. **3**J is a cross-sectional view of a SiC device showing another preferred embodiment along A**2**-A**2'** line of FIG. **3**D according to the present invention.

[0040] FIG. **3**K is a cross-sectional view of a SiC device showing another preferred embodiment along A**2**-A**2'** line of FIG. **3**D according to the present invention.

[0041] FIG. **4**A is a cross-sectional view of another preferred embodiment according to the present invention.

[0042] FIG. **4**B is a cross-sectional view of another preferred embodiment according to the present invention.

[0043] FIG. **4**C is a cross-sectional view of another preferred embodiment according to the present invention.

[0044] FIG. **4**D is a cross-sectional view of another preferred embodiment according to the present invention.

[0045] FIG. **4**E is a cross-sectional view of another preferred embodiment according to the present invention.

[0046] FIG. **4**F is a cross-sectional view of another preferred embodiment according to the present invention.

[0047] FIG. **4**G is a cross-sectional view of another preferred embodiment according to the present invention.

[0048] FIG. **4**H is a cross-sectional view of another preferred embodiment according to the present invention.

[0049] FIG. **5**A is a cross-sectional view of another preferred embodiment integrated with an embedded JBSD wherein doping concentration variations of the P type regions (from p body region to BPS2 region) and the N type regions (from JFET1 region to N+ substrate) are depicted separately along the vertical direction according to the present invention.

[0050] FIG. **5**B is a cross-sectional view of a SiC device showing another preferred embodiment along A**2**-A**2'** line of FIG. **3**D according to the present invention.

[0051] FIG. **5**C is a cross-sectional view of a SiC device showing another preferred embodiment along A**2**-A**2'** line of FIG. **3**D according to the present invention.

[0052] FIG. **5**D is a cross-sectional view of another preferred embodiment according to the present invention.

[0053] FIG. **6**A is a cross-sectional view of another preferred embodiment wherein doping concentration variations of the P type regions (from p body region to BPS2 region) and the N type regions (from JFET1 region to N+ substrate) are depicted separately along the vertical direction according to the present invention.

[0054] FIG. **6**B is a cross-sectional view of a SiC device showing another preferred embodiment along A**2**-A**2'** line of FIG. **3**D according to the present invention.

[0055] FIG. **6**C is a cross-sectional view of a SiC device showing another preferred embodiment along A**2**-A**2'** line of FIG. **3**D according to the present invention.

[0056] FIG. **6**D is a cross-sectional view of another preferred embodiment according to the present invention.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0057] In the following Detailed Description, reference is made to the accompanying drawings, which forms a part thereof, and in which is shown by way of illustration specific embodiments in which the invention may be practiced. In this regard, directional terminology, such as "top", "bottom", "front", "back", etc., is used with reference to the orientation of the Figure(s) being described. Because components of embodiments can be positioned in a number of different orientations, the directional terminology is used for purpose of illustration and is in no way limiting. It is to be understood that other embodiments may be utilized and structural or logical changes may be made without departing from the scope of the present invention. The following detailed description, therefore, is not to be taken in a limiting sense, and the scope of the present invention is defined by the appended claims. It is to be understood that the features of the various exemplary embodiments described herein may be combined with each other, unless specifically noted otherwise.

[0058] Please refer to FIG. **2**A for a preferred embodiment of this invention with a new and improved device structure wherein doping concentration variations of the P type regions (from p body region to BPS2 region) and the N type regions (from JFET1 region to N+ substrate) are depicted separately along the vertical direction. The preferred embodiment comprises a SiC device formed on an N+ type SiC substrate **201** with a less doped N type SiC epitaxial layer **202** extending thereon, wherein the N+ substrate **201** is coated with a back metal **220** on the rear side as a drain metal. The p body regions **210** and n+ source regions **211** are formed at a top portion of the N type SiC epitaxial layer **202**, wherein the n+ source regions **211** are encompassed in the p body regions **210** and a first type JFET (JFET1, as illustrated) region **207** is formed between the two adjacent p body regions **210** with a doping concentration D_{JFET1}, wherein D_{JFET1} is higher than a doping concentration D_N of the N type epitaxial layer **202**, that is $D_{JFET1}>D_N$. An N-drift (ND, as illustrated) region **203** of the first conductivity type is formed between the N+ substrate **201** and the JFET1 region **207**. Inside the N type epitaxial layer **202**, multiple stepped buried P-shield (MSBPS, as illustrated) regions of the second conductivity type are formed adjoining the body region **210** including a bottom second BPS (BPS2, as illustrated) region **228** with a doping concentration D_{BPS2} and a top first BPS (BPS1, as illustrated) region **218** above the BPS2 region **228** with a doping concentration D_{BPS1}, wherein $D_{BPS1}<D_{BPS2}$. As a doping concentration D_p of the p body region **210** is lower than the D_{BPS1}, the relationship among the doping concentrations of the P type regions is $D_p<D_{BPS1}<D_{BPS2}$. A p+ heavily doped region **214** is formed in one side of the p body region **210** above the MSBPS regions opposite to the JFET1 region **207**. Meanwhile, multiple stepped JFET (MSJ, as illustrated) regions of the first conductivity type are formed adjoining the JFET1 region **207** sandwiched between the two adjacent MSBPS regions, comprising a second type JFET (JFET2, as illustrated) region **217** with a width W_{J2} and a doping concentration D_{JFET2} and a third type JFET (JFET3, as illustrated) region **227** below the JFET2 region **227** with a width W_{J3} and a doping concentration D_{JFET3}, wherein $W_{J2}>W_{J3}$ and $D_{JFET2}<D_{JFET3}$, while W_{J2} is less than a width W_{J1} of the JFET1 region **207** and D_{JFET2} is higher than a doping concentration D_{JFET1} of the JFET1 region **207**, the relationships among widths and doping concentrations of the three type JFET regions are $W_{J1}>W_{J2}>W_{J3}$ and $D_{JFET1}<D_{JFET2}<D_{JFET3}$. Moreover, a first saturation current pinching (1st SCP, as illustrated) structure **213** comprising a JFET2 region **217** sandwiched between the two top BPS1 regions **218** and a second saturation current pinching (2nd SCP, as illustrated) structure **223** comprising a JFET3 region **227** sandwiched between the two bottom BPS2 regions **228** are formed to clamp the voltage, thus limiting the saturation current in a forward conduction stage for the short-circuit capability improvement. The doping concentration profile of each region of the MSBPS and the MSJ regions can be uniform (as shown by the solid line in FIG. **2**A) or non-uniform with a peak doping concentration (as shown by the dashed line in FIG. **2**A) in each region depending on the formation method of using epitaxial growth method for the solid line or ion implantation method for the dashed line.

The doping concentration of each region of the MSBPS regions and MSJ regions specified herein or thereafter is defined by its maximum value. A doped poly-silicon layer padded by a gate oxide **209** is formed on the N type epitaxial layer **202** as a plan gate electrode **205** covering the top surfaces of the p body regions **210**, the n+ source regions **211** and the JFET1 regions **207**, wherein the gate oxide **209** has a thick oxide in its central portion. An interlayer dielectric film **221** is then stacked on the planar gate electrode **205**, and a source metal **212** is formed onto the interlayer dielectric film **221**, the n+ source regions **211** and the p regions **210**. The MSBPS regions, the n+ source regions **211** and the p body regions **210** are further shorted together to the source metal **212**.

[0059] Please refer to FIG. **2**B for another preferred embodiment of this invention with a new and improved device structure wherein doping concentration variations of the P type regions (from p body region to BPS2 region) and the N type regions (from JFET1 region to N+ substrate) are depicted separately along the vertical direction. The SiC device has a similar structure to FIG. **2**A, except that the present invention further comprises a current spreading layer (CSL, as illustrated) **204'** of the first conductivity type below the MSBPS and MSJ regions with a doping concentration higher than that of the N type epitaxial layer **202'** for the on-resistance reduction.

[0060] Please refer to FIG. **2**C for another preferred embodiment of this invention with a new and improved device structure wherein doping concentration variations of the P type regions (from p body region to BPS2 region) and the N type regions (from JFET1 region to N+ substrate) are depicted separately along the vertical direction. The SiC device has a similar structure to FIG. **2**B, except that the MSBPS regions in the present invention comprises a bottom third BPS (BPS3, as illustrated) region **238"** with a doping concentration D_{BPS3}, a middle second BPS (BPS2, as illustrated) region **228"** with a doping concentration D_{BPS2} and a top first BPS (BPS1, as illustrated) region **218"** above the middle BPS2 region **228"** with a doping concentration D_{BPS1}, wherein $D_{BPS1}<D_{BPS2}<D_{BPS3}$. Moreover, MSJ regions in the present invention comprises a second type JFET (JFET2, as illustrated) region **217"** with a width W_{J2} and a doping concentration D_{JFET2}, a third type JFET (JFET3, as illustrated) region **227"** below the JFET2 region **217"** with a width W_{J3} and a doping concentration D_{JFET3} and a fourth type JFET (JFET4, as illustrated) region **237"** below the JFET3 region **227"** with a width W_{J4} and a doping concentration D_{JFET4}, wherein $W_{J2}>W_{J3}>W_{J4}$ and $D_{JFET2}<D_{JFET3}<D_{JFET4}$.

[0061] Please refer to FIG. **2**D for another preferred embodiment of this invention with a new and improved device structure. The SiC device has a similar structure to FIG. **2**A, except that the present invention further comprises P column (PC, as illustrated) regions **216'"**of the second conductivity type formed adjoining bottom surfaces of the BPS2 regions **228'"** above the N+ substrate **201'"**. A super junction (SJ, as illustrated) structure is thus generated by the N type epitaxial layer **202'"** and the PC region **216'"**.

[0062] Please refer to FIG. **2**E for another preferred embodiment of this invention with a new and improved device structure. The SiC device has a similar structure to FIG. **2**D, except that the present invention further comprises an additional N buffer layer (Nb, as illustrated) **222""** with a resistivity Rb sandwiched between the N+ substrate **201""** and the PC regions **216""**, wherein Rb is higher than a resistivity R of the N type epitaxial layer **202""**.

[0063] Please refer to FIG. **2**F for another preferred embodiment of this invention with a new and improved device structure. The SiC device has a similar structure to FIG. **2**D, except for the different substrate. In this invention, the SiC device is formed on a P+ substrate **201"'"**, and the invention in FIG. **2**F further comprises an N buffer layer (Nb, as illustrated) **222"'"** with a resistivity Rb sandwiched between the P+ substrate **201"'"** and the PC regions **216"'"**, wherein Rb is lower than a resistivity R of the N type epitaxial layer **202"'"**.

[0064] Please refer to FIG. **2**G for another preferred embodiment of this invention with a new and improved device structure. The SiC device has a similar structure to FIG. **2**F, except that, the SiC device in FIG. **2**G further comprises a plurality of heavily doped N+ regions **240"'"** in the P+ substrate **201"'"** to form a plurality of alternating P+ and N+ regions in the substrate.

[0065] Please refer to FIG. **3**A for a top view of a preferred embodiment for a SiC device comprising a plurality of unit cells with each unit cell in an active area. A first BPS region of the second conductivity type is formed including a left side portion BPS1L **318**, a central portion BPS1C **308**, a right side portion BPS1R **328** and connection portions BPS1G, wherein BPS1G comprises a left side portion BPS1GL **319** and a right side portion BPS1GR **329** and connect the BPS1C **308** together with BPS1L **318** and BPS1R **328** to a source metal through source contact **335** surrounded by a gate poly silicon layer **305** as a gate electrode. A second type JFET (JFET2) region is formed having a left side portion JFET2L **317** disposed between the BPS1L **318** and the BPS1C **308**, and a right side portion JFET2R **327** disposed between the BPS1C **308** and the BPS1R **328**.

[0066] Please refer to FIG. **3**B for a cross-sectional view showing a preferred A**1**-A**1'** cross section of FIG. **3**A according to the present invention. The preferred embodiment comprises a SiC device formed on an N+ type SiC substrate **301'** with a less doped N type SiC epitaxial layer **302'** extending thereon, wherein the N+ substrate **301'** is coated with a back metal **320'** on the rear side as a drain metal. The p body regions **310'** and n+ source regions **311'** are formed at a top portion of the N type SiC epitaxial layer **302'**, wherein the n+ source regions **311'** are encompassed in the p body regions **310'** and a first type JFET (JFET1, as illustrated) region **307'** is formed between the two adjacent p body regions **310'**. Inside the N type epitaxial layer **302'**, a first BPS region of the second conductivity type is formed including a left side portion BPS1L **318'**, a center portion BPS1C **308'** and a right side portion BPS1R **328'**, and the BPS1L **318'** is shifted from the p body region **310'** toward the center of the gate electrode **305'** by a distance LD_{PBPS}, wherein $LD_{PBPS}>0$. A second type JFET (JFET2) region is formed having a left side portion JFET2L **317'** and a right side portion JFET2R **327'**. Therefore, a p+ heavily doped region **314'** is formed in one side of the p body region **310'** above the BPS1L region **318'** and BPS1R region **328'** opposite to the JFET1 region **307'**. A dual saturation current pinching (SCP) structure is formed comprising a left side SCP (SCPL, as illustrated) structure **313'** and a right side SCP (SCPR, as illustrated) structure **323'** to clamp the voltage, wherein the SCPL structure comprises a JFET2L region **317'** sandwiched between the BPS1L region **318'** and

the BPS1C region **308'**, and the SCPR structure **323'** comprises a JFET2R region **327'** sandwiched between the BPS1C region **308'** and the BPS1R region **328'**. A doped poly-silicon layer padded by a gate oxide **309'** is formed on the N type epitaxial layer **302'** as a planar gate electrode **305'** covering the top surfaces of the p body regions **310'**, the n+ source regions **311'** and the JFET1 region **307'**, wherein the gate oxide **309'** has a thick oxide in its central portion. An interlayer dielectric film **321'** is then stacked on the planar gate electrode **305'**, and a source metal **312'** is formed onto the interlayer dielectric film **321'**, the n+ source regions **311'** and the p body regions **310'**. The BPS1L **318'**, the BPS1R **328'**, the n+ source regions **311'** and the p body regions **310'** are further shorted together to the source metal **312'**.

[0067] Please refer to FIG. **3**C for a cross-sectional view showing a preferred B**1**-B**1'** cross section of FIG. **3**A according to the present invention. The SiC device has a similar structure to FIG. **3**B, except that the first BPS region in this invention comprises a left side portion BPS1L **318"**, a center portion BPS1C **308"**, a right side portion BPS1R **328"** and connection portions BPS1G, wherein BPS1G comprises BPS1GL **319"** and BPS1GR **329"** and connects the BPS1C **308"** together with the BPS1L **318"** and the BPS1R **328"** to a source metal **312"** through p body regions **310"**. Moreover, the JFET2 region in FIG. **3**B doesn't exist in this cross section.

[0068] Please refer to FIG. **3**D for a top view of a preferred embodiment for a SiC device comprising a plurality of unit cells with each unit cell in an active area. The SiC device has a similar structure to FIG. **3**A, except that the first BPS region of the second conductivity type in the present invention comprises a left side portion BPS1L **318'"**, a center portion BPS1C **308'"**, a right side portion BPS1R **328'"** and a right side connection portion BPSGR **329'"**.

[0069] Please refer to FIG. **3**E for a cross-sectional view showing a preferred A**2**-A**2'** cross section of FIG. **3**D according to the present invention, and the SiC device has a same structure as FIG. **3**B.

[0070] Please refer to FIG. **3**F for a cross-sectional view showing a preferred B**2**-B**2'** cross section of FIG. **3**D according to the present invention. The SiC device has a similar structure to FIG. **3**C, except that the first BPS region in this invention comprises a left side portion BPS1L **318''''**, a center portion BPS1C **308''''**, a right side portion BPS1R **328''''** and connection portion BPSGR **329''''**, and BPSGR **329"** connects the BPS1C **308''''** together with BPS1R **328''''** to a source metal **312''''**. Moreover, the present invention further comprises a second type JFET (JFET2) region having a left side portion JFET2L **317''''** disposed between the BPS1L **318''''** and the BPS1C **308''''**, and a single SCPL structure **313''''** comprising a JFET2L **317''''** is formed in this invention.

[0071] Please refer to FIG. **3**G for a cross-sectional view showing another preferred A**2**-A**2'** cross section of FIG. **3**D according to the present invention. The SiC device has a similar structure to FIG. **3**E, except that the present invention further comprises a current spreading layer (CSL, as illustrated) **1304** of the first conductivity type below the first BPS region and the second type JFET region with a doping concentration higher than that of the N type epitaxial layer **1302**, wherein the first BPS region comprises a left side portion BPS1L **1318**, a center portion BPS1C **1308** and a right side portion BPS1R **1328**, and the second type JFET region comprises a left side portion JFET2L **1317** and a right side portion JFET2R **1327**.

[0072] Please refer to FIG. **3**H for a cross-sectional view showing another preferred A**2**-A**2'** cross section of FIG. **3**D according to the present invention. The SiC device has a similar structure to FIG. **3**G, except that in the present invention, the $LD_{PBPS}<0$.

[0073] Please refer to FIG. **3**I for a cross-sectional view showing another preferred A**2**-A**2'** cross section of FIG. **3**D according to the present invention. The SiC device has a similar structure to FIG. **3**E, except that the present invention further comprises P column (PC, as illustrated) regions **1316"**of the second conductivity type formed adjoining bottom surfaces the BPS1L region **1318"** and the BPS1R region **1328"** above the N+ substrate **1301"**. A super junction (SJ, as illustrated) structure is thus generated by the N type epitaxial layer **1302"** and the PC region **1316"**.

[0074] Please refer to FIG. **3**J for a cross-sectional view showing another preferred A**2**-A**2'** cross section of FIG. **3**D according to the present invention. The SiC device has a similar structure to FIG. **31**, except that the present invention further comprises an additional N buffer layer (Nb, as illustrated) **1322'"** with a resistivity Rb sandwiched between the N+ substrate **1301'"** and the PC regions **1316'"**, wherein Rb is higher than a resistivity R of the N type epitaxial layer **1302'"**.

[0075] Please refer to FIG. **3**K for a cross-sectional view showing another preferred A**2**-A**2'** cross section of FIG. **3**D according to the present invention. The SiC device has a similar structure to FIG. **3**I, except for the different substrate. In this invention, the SiC device is formed on a P+ substrate **1301''''**, and the invention in FIG. **3**K further comprises an N buffer layer (Nb, as illustrated) **1322''''** with a resistivity Rb sandwiched between the P+substrate **1301''''** and the PC regions **1316''''**, wherein Rb is lower than a resistivity R of the N type epitaxial layer **1302"41** .

[0076] Please refer to FIG. **4**A for another preferred embodiment of this invention with a new and improved device structure. The SiC device has a similar structure to FIG. **3**B, except that the present invention in FIG. **4**A has a cross sectional width twice of that in FIG. **3**B, and further comprises a second BPS (BPS2, as illustrated) region **448** of the second conductivity formed below BPS1L region **418**, BPS1R region **428** and BPS1 region **438**, and a third type JFET (JFET3, as illustrated) region **437** formed between the two adjacent BPS2 regions **448**, wherein the BPS2 region **448** has a doping concentration higher than that of the first type BPS region, and having a width W_{BPS2} greater than a width W_{BPS1} of the BPS1 region **438**. The BPS1 regions **438**, the BPS2 region **448**, the n+ source regions **411** and the p body regions **410** are further shorted together to the source metal **412** through a source contact **435**. Moreover, a third saturation current pinching (3rd SCP, as illustrated) structure **433** comprising a JFET3 region **437** sandwiched between the BPS2 regions **448** is formed to further clamp the voltage.

[0077] Please refer to FIG. **4**B for another preferred embodiment of this invention with a new and improved device structure. The SiC device has a similar structure to FIG. **3**B, except that the present invention in FIG. **4**B further comprises a second BPS region **448'** of the second conductivity including a left portion BPS2L **438'** and a right portion BPS2R **448'** formed below the BPS1L **418'** and the BPS1R **428'**, respectively, wherein the BPS2 region has a doping

concentration higher than that of the first type BPS region. Moreover, a third saturation current pinching (3rd SCP, as illustrated) structure **433'** comprising a JFET3 region **437'** sandwiched between the BPS2L **438'** and the BPS2R **448'** is formed to further clamp the voltage. The present invention further comprises a current spreading layer (CSL, as illustrated) **404'** of the first conductivity type below the MSBPS regions with a doping concentration higher than a doping concentration of the N type epitaxial layer **402'**.

[0078] Please refer to FIG. **4**C for another preferred embodiment of this invention with a new and improved device structure. The SiC device has a similar structure to FIG. **4**B, except that the present invention in FIG. **4**C further comprises a third BPS region of the second conductivity including a left portion BPS3L **458"** and a right portion BPS3R **468"** formed below the BPS2L **438"** and BPS2R **448'**, respectively. Moreover, a fourth saturation current pinching (4th SCP, as illustrated) structure **443"** comprising a fourth type JFET (JFET4, as illustrated) region **447"** sandwiched between the BPS3L **458"** and the BPS3R **468"** is formed to further clamp the voltage, and the current spreading layer (CSL, as illustrated) **404'** in FIG. **4**B doesn't exist in the present invention.

[0079] Please refer to FIG. **4**D for another preferred embodiment of this invention with a new and improved device structure. The SiC device has a similar structure to FIG. **4**C, except that the present invention in FIG. **4**D further comprises a current spreading layer (CSL, as illustrated) **404'"** of the first conductivity type below the MSBPS regions with a doping concentration higher than a doping concentration of the N type epitaxial layer **402'"**.

[0080] Please refer to FIG. **4**E for another preferred embodiment of this invention with a new and improved device structure. The SiC device has a similar structure to FIG. **4**B, except that the current spreading layer (CSL, as illustrated) **404'** in FIG. **4**B doesn't exist in the present invention, and the SiC device in FIG. **4**E further comprises P column (PC, as illustrated) regions **416""** of the second conductivity type formed adjoining bottom surfaces the BPS2L **438""** and the BPS2R **448""** above the N+ substrate **401""**. A super junction (SJ, as illustrated) structure is thus generated by the N type epitaxial layer **402""** and the PC region **416""**.

[0081] Please refer to FIG. **4**F for another preferred embodiment of this invention with a new and improved device structure. The SiC device has a similar structure to FIG. **4**E, except that the present invention further comprises an additional N buffer layer (Nb, as illustrated) **1422** with a resistivity Rb sandwiched between the N+ substrate **1401** and the PC region **1416**, wherein Rb is higher than a resistivity R of the N type epitaxial layer **1402**.

[0082] Please refer to FIG. **4**G for another preferred embodiment of this invention with a new and improved device structure. The SiC device has a similar structure to FIG. **4**E, except for the different substrate. In this invention, the SiC device is formed on a P+substrate **1401'**, and further comprises an N buffer layer (Nb, as illustrated) **1422'** with a resistivity Rb sandwiched between the P+substrate **1401'** and the PC region **1416'**, wherein Rb is lower than a resistivity R of the N type epitaxial layer **1402'**.

[0083] Please refer to FIG. **4**H for another preferred embodiment of this invention with a new and improved device structure. The SiC device has a similar structure to FIG. **4**G, except that, the SiC device in FIG. **4**H further comprises a plurality of heavily doped N+ regions **1440"** in the P+ substrate **1401"** to form a plurality of alternating P+ and N+ regions in the substrate.

[0084] Please refer to FIG. **5**A for another preferred embodiment of this invention with a new and improved device structure wherein doping concentration variations of the P type regions (from p body region to BPS2 region) and the N type regions (from JFET1 region to N+ substrate) are depicted separately along the vertical direction. The SiC device has a similar structure to FIG. **2**B, except that the planar electrode in the present invention is a split gate electrode having two gate electrodes **505** padded by a gate oxide **509**, and two p+ heavily doped regions **524** of a second conductivity are formed within the JFET1 region **507**, wherein the two p+ regions **524** are spaced apart from the p body region **510**. The SiC device further comprises a Junction barrier Schottky diode (JBSD, as illustrated) embedded between the two adjacent gate electrodes **505** by making a Schottky contact with the JFET1 region **507** between the two p+ regions **524** within the JFET1 region **507**. Therefore, a Junction Barrier Schottky diode (JBSD, as illustrated) is integrated with the SiC MOSFET in a location between the two adjacent split gate electrodes **505** for the reverse conduction switching loss reduction.

[0085] Please refer to FIG. **5**B for a cross-sectional view showing another preferred A**2**-A**2'** cross section of FIG. **3**A according to the present invention. The SiC device has a similar structure to FIG. **3**B, except that the planar electrode in the present invention is a split gate electrode having two gate electrodes **505'** padded by a gate oxide **509'**, and two p+ heavily doped regions **524'** of a second conductivity type are formed within the JFET1 region **507'**, wherein the two p+ regions **524'** are spaced apart from the p body region **510'**. Moreover, the SiC device further comprises a Junction barrier Schottky diode (JBSD, as illustrated) embedded between the two adjacent gate electrodes **505"** by making a Schottky contact with the JFET1 region **507'** between the two p+regions **524'** within the JFET1 region **507'**. Therefore, a Junction Barrier Schottky diode (JBSD, as illustrated) is integrated with the SiC MOSFET in a location between the two adjacent split gate electrodes **505'** for the reverse conduction switching loss reduction.

[0086] Please refer to FIG. **5**C for a cross-sectional view showing another preferred A**2**-A**2'** cross section of FIG. **3**A according to the present invention. The SiC device has a similar structure to FIG. **5**B, except that in the present invention, the $LD_{PBPS}<0$.

[0087] Please refer to FIG. **5**D for another preferred embodiment of this invention with a new and improved device structure. The SiC device has a similar structure to FIG. **4**A, except that the planar electrode in the present invention is a split gate electrode having two gate electrodes **505'"** padded by a gate oxide **509'"**, and two p+ heavily doped regions **524'"** are formed within the JFET1 region **507'"**, wherein the two p+ regions **524'"** are spaced apart from the p body region **510'"**. Moreover, the SiC device further comprises a Junction barrier Schottky diode (JBSD, as illustrated) embedded between the two adjacent gate electrodes **505'"** by making a Schottky contact with the JFET1 region **507'"** between the two p+ regions **524'"** within the JFET1 region **507'"**. Therefore, a Junction Barrier Schottky diode (JBSD, as illustrated) is integrated with the

SiC MOSFET in a location between the two adjacent split-gate electrodes **505'''** for the reverse conduction switching loss reduction.

[0088] Please refer to FIG. **6**A for another preferred embodiment of this invention with a new and improved device structure wherein doping concentration variations of the P type regions (from p body region to BPS2 region) and the N type regions (from JFET1 region to N+ substrate) are depicted separately along the vertical direction. The SiC device has a similar structure to FIG. **5**A, except that the two p+ heavily doped regions **524** in FIG. **5**A are replaced by a p+ heavily doped regions **624** within the JFET1 region **607** in the present invention, wherein the p+ region **624** is spaced apart from the p body region **610**. The SiC device further comprises a P/N (P-type/N-type) junction diode embedded between the two split gate electrodes **605** by making a p+ region **624** of a second conductivity type spaced apart from the p body region **610'** within the JFET1 region **607** of the first conductivity type for gate oxide electric field reduction.

[0089] Please refer to FIG. **6**B for a cross-sectional view showing another preferred A**2**-A**2'** cross section of FIG. **3**A according to the present invention. The SiC device has a similar structure to FIG. **5**B, except that the two p+ heavily doped regions **524'** in FIG. **5**A are replaced by a p+ heavily doped regions **624'** within the JFET1 region **607'** in the present invention, wherein the p+region **624'** is spaced apart from the body region **610'**. The SiC device further comprises a P/N (P-type/N-type) junction diode embedded between the two split gate electrodes **605'** by making a p+ region **624'** of a second conductivity type spaced apart from the p body region **610'** within the JFET1 region **607'** of the first conductivity type.

[0090] Please refer to FIG. **6**C for a cross-sectional view showing another preferred A**2**-A**2'** cross section of FIG. **3**A according to the present invention. The SiC device has a similar structure to FIG. **6**B, except that in the present invention, the $LD_{PBPS}<0$.

[0091] Please refer to FIG. **6**D for another preferred embodiment of this invention with a new and improved device structure. The SiC device has a similar structure to FIG. **5**D, except that the two p+ heavily doped regions **524'''** in FIG. **5**A are replaced by a p+ heavily doped regions **624'''** within the JFET1 region **607'''** in the present invention, wherein the p+ region **624'''** is spaced apart from the body region **610'''**. The SiC device further comprises a P/N (P-type/N-type) junction diode embedded between the two split gate electrodes **605'''** by making a p+ region **624'''** of a second conductivity type spaced apart from the p body region **610'''** within the JFET1 region **607'''** of the first conductivity type.

[0092] Although the present invention has been described in terms of the presently preferred embodiments, it is to be understood that such disclosure is not to be interpreted as limiting. Various alternations and modifications will no doubt become apparent to those skilled in the art after reading the above disclosure. Accordingly, it is intended that the appended claims be interpreted as covering all alternations and modifications as fall within the true spirit and scope of the invention.

What is claimed is:

1. A silicon carbide (SiC) device comprising a plurality of unit cells with each unit cell in an active area, comprising:

an epitaxial layer grown on a substrate;

a source region of a first conductivity type formed at a top portion of said epitaxial layer and encompassed in a body region of a second conductivity type;

a planar gate electrode made of a doped poly-silicon layer padded by a gate oxide;

a first type Junction Field Effect Transistor (JFET1) region of said first conductivity type formed between two adjacent said body regions;

multiple stepped buried P-shield (MSBPS) regions of said second conductivity type adjoining said body region with different doping concentrations increasing stepwise in a direction from said body region toward said substrate;

multiple stepped JFET (MSJ) regions of said first conductivity type sandwiched between two adjacent said MSBPS regions with different widths decreasing stepwise in a direction from said JFET1 region toward said substrate;

said MSJ regions with different doping concentrations increasing stepwise in a direction from said JFET1 region toward to substrate; and

an N-drift (ND) region of said first conductivity type formed between said substrate and said first type JFET1 region.

2. The SiC device of claim **1**, further comprising a current spreading layer (CSL) of said first conductivity type below said MSBPS and said MSJ regions with a doping concentration higher than that of said ND region.

3. The SiC device of claim **1**, wherein said planar gate electrode is a single gate electrode padded with said gate oxide having a thick oxide in its central portion.

4. The SiC device of claim **1**, wherein said planar gate electrode is a split gate electrode having two gate electrodes in said each unit cell, and further comprising a Junction barrier Schottky diode (JBSD) embedded between two adjacent said gate electrodes by making a Schottky contact with said first type JFET (JFET1) region between two p+ regions of said second conductivity type within said JFET1 region, wherein said two p+ regions are spaced apart from said body region.

5. The SiC device of claim **1**, wherein said planar gate electrode is a split gate electrode having two gate electrodes in said each unit cell, and further comprising a P/N (P-type/N-type) junction diode embedded between two adjacent said gate electrodes by making a p+ region of said second conductivity type within said JFET1 region, wherein said p+ region is spaced apart from said body region.

6. The SiC device of claim **1**, wherein said MSBPS regions comprise at least two buried P-shield (BPS) regions including a top first BPS (BPS1) region with a doping concentration D_{BPS1} and a bottom second BPS (BPS2) region with a doping concentration D_{BPS2}, wherein said $D_{BPS1}<D_{BPS2}$; said MSJ regions comprise at least two JFET regions including a second type JFET (JFET2) region with a width W_{J2} and a doping concentration D_{JFET2} and a third type JFET (JFET3) region below said JFET2 region with a width W_{J3} and a doping concentration D_{JFET3}, wherein said $W_{J2}>W_{J3}$ and said $D_{JFET2}<D_{JFET3}$, while W_{J2} is less than a width W_{J1} of said JFET1 region and D_{JFET2} is higher than a doping concentration D_{JFET1} of said JFET1 region, the relationships among widths and doping concentrations of said three type JFET regions are said $W_{J1}>W_{J2}>W_{J3}$ and said $D_{JFET1}<D_{JFET2}<D_{JFET3}$; and said MSBPS and said MSJ regions have an uniform doping concentration profile or a non-uniform doping profile with a peak doping concentration in each region.

7. The SiC device of claim **1**, further comprising a super junction structure comprising a P column region of said second conductivity type disposed above said substrate.

8. The SiC device of claim **7**, wherein said substrate has said first conductivity type and further comprises a buffer layer of said first conductivity type with a resistivity Rb sandwiched between said substrate and said epitaxial layer with a resistivity R, wherein said R<Rb.

9. The SiC device of claim **7**, wherein said substrate has said second conductivity type and further comprises a buffer layer of said first conductivity type with a resistivity Rb sandwiched between said substrate and said epitaxial layer with a resistivity R, wherein said R>Rb.

10. The SiC device of claim **7**, wherein said substrate has said second conductivity type, further comprises a buffer layer of said first conductivity type formed sandwiched between said substrate and said epitaxial layer; a plurality of heavily doped regions of said first conductivity type in said substrate to form a plurality of alternating P+ and N+ regions in said substrate.

11. A SiC device comprising a plurality of unit cells with each unit cell in an active area, comprising:

an epitaxial layer grown on a substrate;

a source region of a first conductivity type formed at a top portion of said epitaxial layer and encompassed in a body region of a second conductivity type;

a planar gate electrode made of a doped poly-silicon layer padded by a gate oxide;

a first type JFET (JFET1) region of said first conductivity type formed between two adjacent said body regions;

a first BPS (BPS1) region of said second conductivity type formed below said body region and said JFET1 region having a left side portion BPS1L, a center portion BPS1C, a right side portion BPS1R and connection portions BPS1G;

said BPS1G connecting said BPS1C together with BPS1L and/or BPS1R to a source metal through said body region;

a second type JFET (JFET2) region having a left side portion JFET2L formed between said BPS1L and BPS1C, and a right side portion JFET2R formed between said BPS1C and said BPS1R;

said JFET2 region has a doping concentration higher than that of said JFET1 region; and

an ND region of said first conductivity type is formed between said substrate and said JFET1 region.

12. The SiC device of claim **11**, wherein said planar gate electrode is a single gate electrode padded with said gate oxide having a thick oxide in its central portion.

13. The SiC device of claim **11**, wherein said planar gate electrode is a split gate electrode having two gate electrodes in said each unit cell, and further comprising a JBSD embedded between two adjacent said gate electrodes by making a Schottky contact with said JFET1 region between two p+ regions of said second conductivity type within said JFET1 region. wherein said two p+ regions are spaced apart from said body region

14. The SiC device of claim **11**, wherein said planar gate electrode is a split gate electrode having two gate electrodes in said each unit cell, and further comprising a P/N (P-type/N-type) junction diode embedded between two adjacent said gate electrodes by making a p+ region of said second conductivity type within said JFET1 region, wherein said p+ region is spaced apart from said body region.

15. The SiC device of claim **11**, further comprising a second BPS (BPS2) region of said second conductivity type formed below said BPS1 region and a third type JFET (JFET3) region formed between two adjacent said BPS2 regions; said BPS2 region having a doping concentration higher than that of said BPS1 region, and having a width greater than that of said BPS1 region.

16. The SiC device of claim **11**, further comprising a CSL below said BPS2 and said JFET3 regions with a doping concentration higher than that of said ND region.

17. The SiC device of claim **11**, further comprising a super junction structure comprising a P column region of said second conductivity type disposed above said substrate.

18. The SiC device of claim **17**, wherein said substrate has said first conductivity type and further comprises a buffer layer of said first conductivity type with a resistivity Rb sandwiched between said substrate and said epitaxial layer with a resistivity R, wherein said R<Rb.

19. The SiC device of claim **17**, wherein said substrate has said second conductivity type and further comprises a buffer layer of said first conductivity type with a resistivity Rb sandwiched between said substrate and said epitaxial layer with a resistivity R, wherein said R>Rb.

20. The SiC device of claim **17**, wherein said substrate has said second conductivity type, further comprises a buffer layer of said first conductivity type formed sandwiched between said substrate and said epitaxial layer; a plurality of heavily doped regions of said first conductivity type in said substrate to form a plurality of alternating P+ and N+ regions in said substrate.

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